Computation of Wilson Loops and Verifying Phase Transition in Bosonic BFSS

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Certificate of Examination

This is to certify that the dissertation titled **Computation of Wilson Loops and Verifying Phase Transition in Bosonic BFSS** submitted by **Raunok Basu** (Reg. No. MS17088) for the partial fulfilment of BS-MS Dual Degree programme of the institute, has been examined by the thesis committee duly appointed by the institute. The committee finds the work done by the candidate satisfactory and recommends that the report be accepted.

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Declaration

The work presented in this dissertation has been carried out by me under the guidance of Dr. Anosh Joseph at the Indian Institute of Science Education and Research (IISER) Mohali.

This work has not been submitted in part or in full for a degree, a diploma, or a fellowship to any other university or institute. Whenever contributions of others are involved, every effort is made to indicate this clearly, with due acknowledgement of collaborative research and discussions. This thesis is a bona fide record of original work done by me, and all sources listed within have been detailed in the bibliography.

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In my capacity as the supervisor of the candidate's project work, I certify that the above statements by the candidate are true to the best of my knowledge.

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Abstract

In this work, we try to understand how the strongly-coupled gauge theory is related to the weakly-coupled dual gravity. In order to understand this, we try to evaluate the Schwarzschild radius of a black hole from the Wilson loop observable in the BFSS Matrix Model. Further, we also seek to understand the confined-deconfined phase transition by evaluating the Polyakov loop operator, which is the order parameter for the transition.

List of Figures

List of Tables

Contents

Chapter 1

Introduction

String theory is one of the leading candidates for the theory of everything as envisioned by various physicists over the years. The theory of everything seeks to unify the four fundamental forces in nature - gravity, weak, strong, and electromagnetic. The starting point of the theory is that one can replace zero-dimensional particles, as is the norm in understanding classical physics, with one-dimensional strings with every distinct particle being a unique vibrational mode of an elementary macroscopic string.

An important feature of string theory is its uniqueness, as the theory lacks adjustable dimensionless parameters. There are five consistent superstring theories: type I, type IIA, type IIB, heterotic $SO(32)$ and heterotic $E_8 \times E_8$. These theories differ based on the types of strings involved (open/closed, oriented/non-oriented, etc). All of these theories exist in 10-dimensional spacetime, so for the theory to be successfully explain the real world phenomena, there must exist some mechanism that would ensure that only four dimensions are visible [Zwiebach 04].

Since all these string theories mentioned above are related by a transformation like S-duality, Edward Witten conjectured in 1995 that they must be part of some bigger theory called M-Theory. It is only a conjecture still and in different parameter regimes, it gives rise to the five string theories and the eleven dimensional supergravity.

A significant achievement of string theory is that the quantum theory of gravity is found among the quantum vibrations of a relativistic string. It is a substantial improvement on the Standard Model, where no evidence of gravity has been experimentally observed yet. String theories have been an active area of research since 1984, when Michael Green and John Schwarz showed that superstrings are not afflicted with those inconsistencies that plague other particle theories in ten dimensions.

String/gauge duality, which originated from the AdS/CFT correspondence, has been investigated intensively over the past decade. Remarkable developments that have been achieved include generalisation to various cases, confirmation by explicit calculations, and applications to multiple branches of physics such as hadron physics and condensed matter physics. From the viewpoint of string theory, the duality enables us to study quantum aspects of gravity, including its non-perturbative effects from the gauge theory side, which is more tractable. In this regard, it is vital to understand how gauge theories capture the information of space-time geometry. On the basis of the duality at finite temperature, it can be shown that a temporal Wilson loop operator in gauge theory is related directly to the Schwarzschild radius [Rey 98], which is a fundamental quantity that characterizes the dual black hole geometry.

1.1 BFSS Matrix Model

The BFSS matrix model was conjectured in 1996 by Banks, Fischler, Susskind and Shenker [Banks 97]. This model is the one-dimensional superstring Yang-Mills theory and this theory acts as a low-energy description of *D0*-branes of Type IIA superstring theory. One of the methods to obtain this action is by dimensionally reducing the $\mathcal{N}=1$ Super Yang-Mills theory in $9+1$ dimensions with gauge group $SU(N)$ to $0+1$ dimensions, which is described in [Tanwar 20]. The action is given by

$$\begin{aligned} S \; &= \frac{1}{g^2}\int_0^\beta dt \;\; \mathrm{Tr}\, \Big[(D_tX_i)^2 - \frac{1}{2}[X_i,X_j]^2 + \imath\Psi_\alpha^T C_{10}\gamma^0_{\alpha\beta} D_t\Psi_\beta \\ &\qquad\qquad - C_{10}\gamma^i_{\alpha\beta}[X_i,\Psi_\beta]\Big], \end{aligned} \tag{1.1}$$

where $D_t = \partial_t - \imath[A(t),\cdot]$; $i,j=1,2,\cdots,9$; $\alpha,\beta=1,2,\cdots,32$; X_i are $N\times N$ traceless, Hermitian matrices; g is the one-dimensional Yang-Mills coupling; $A(t)$ is the one-dimensional gauge field; Ψ is 32-component Majorana-Weyl fermions; with each fermion being an $N\times N$ traceless, Hermitian matrix; C_{10} is the charge conjugation matrix in 10 dimensions; and γ^i are the gamma matrices in 10 dimensions.

1.2 Motivations for Studying the BFSS Model

One of the principal aims of string theory is to study the black hole states. Ever since the pioneering work of Hawking and Bekenstein, which showed that black holes possess both entropy and temperature, there was a need for the statistical interpretation of black holes. String theory provides such a mechanism as black holes can be assembled using various types of *D*-branes and strings. It can also be studied non-perturbatively by using the BFSS matrix model.It is believed that the finite temperature states of the BFSS model are related to the states of type IIA supergravity [Klebanov 98]. The connection between the BFSS matrix model and the black hole states has also been discussed in [Kabat 01a] and [Kabat 01b]. In these two references, the authors calculated the entropy of a black hole in strongly coupled quantum mechanics beginning with the Bekenstein-Hawking entropy of a ten-dimensional non-extremal back hole. The free energy of the black hole in terms of the parameters of the gauge theory (here the BFSS model) can be written as

$$\frac{F}{T} = -4.115N^2\left(\frac{T^2}{g_{YM}^2 N}\right)^{\frac{3}{5}}. \tag{1.2}$$

The above formula is of interest for a couple of reasons. Firstly, its dependence on the 't Hooft coupling. The limit $N \to \infty$ corresponds to the region where supergravity is valid and therefore it is the appropriate limit to look for black holes in the BFSS model. The second reason is that the free energy depends on N^2, which is the expected behaviour of a gauge theory in the deconfined phase.

It would be interesting to reproduce this formula from the matrix model. To get this behaviour in the matrix model, one must look at the strong coupling regime, which is not accessible through perturbation theory. That is, in a regime where it is highly impossible to carry out an analytical calculation to derive this expression. Thus we are left with numerical simulations as these techniques do not have any such restrictions.

1.3 Schwarzschild Geometry of a Black Hole

The Schwarzschild geometry depicts, to an excellent approximation, the empty space around a spherical source of curvature like a spherical star.

It was first discovered as a solution to Einstein's equations in vacuum by Karl Schwarzschild in 1916 [Schwarzschild 16]. The Schwarzschild metric ($G = c = 1$) in 3-dimensional space, is given as

$$ds^2 = -\left(1-\frac{2M}{r}\right)dt^2 + \left(1-\frac{2M}{r}\right)^{-1}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{1.3}$$

In this geometry, we can see that there are singularities at $r = 0$ and $r = 2M$ where M denotes the mass of the spherical object. The latter, known as the Schwarzschild radius, is the characteristic length scale of curvature in Schwarzschild geometry. Usually, for stars, this radius is very small, which means that it lies within the interior of the star. However, this radius becomes significant after the gravitational collapse of a star to form a black hole. This radius corresponds to a regular singularity in the geometry of non-rotating, uncharged black holes and leads to the observation that outside the Schwarzschild radius, no information is obtainable about the interior of the black hole.

According to the cosmic censorship conjecture, the formation of such a singularity (event horizon) is necessary to hide the essential singularity at $r = 0$. It was found that isolated black holes emit Hawking radiation like a black body at the temperature known as the Hawking temperature. This suggests that black holes have entropy, and with the advent of string theory, consistent models for black hole states have emerged.

1.4 Thermal Phase Transition

For a system with density of states that grows exponentially, i.e.,

$$\rho(E) \approx \exp(\beta E) \tag{1.4}$$

there exists an upper limit of temperature. Above this temperature the partition function diverges

$$\lim_{T\to T_H^-} \mathrm{Tr}\, \exp(-\beta E) \to \infty. \tag{1.5}$$

Above this temperature, it becomes impossible for a system with exponentially large number of particles, all with the same energy E and a finite size, to exist. However, it can be made to exist if we keep N large but finite. Performing this breaks the exponential growth in the asymptotic density of states at some large but finite energy value. For temperature greater than T_H, the entropy and the energy are dominated by the states at and above the cutoff scale and the free energy jumps from $O(1)$ to $O(N^2)$. Thus the system undergoes a transition from the "confined" to the "deconfined" phase. This type of transition is found in large N gauge theories such as the weakly-coupled Yang-Mills theory and thus also must be found in the BFSS matrix model [Semenoff 04].

This phase transition in the BFSS model is due to the breakdown of the centre symmetry i.e., $A(t) \to A(t) + kI_n$. The order parameter for this symmetry breaking is the Polyakov loop [Polyakov 78, Kawahara 07]. This phase transition is a second-order phase transition. A more detailed analysis has revealed that there are, in fact, two transitions. The main goal of these studies has been to compare the low-temperature regime of the model to the holographically dual black hole geometry. Polyakov loop is defined as the trace of the holonomy of the gauge field around the finite temperature Euclidean time circle.

$$P = \frac{1}{N}\mathrm{Tr}\Big(\mathscr{P} e^{\imath \oint A(t)dt}\Big). \tag{1.6}$$

This operator is gauge-invariant and is zero in the confined phase and becomes non-zero in the deconfined phase. This is because the Polyakov loop is a unitary matrix and its eigenvalues are uniformly distributed in the unit circle in the confined phase and clumps towards a certain value in the deconfined phase.

Chapter 2

Monte Carlo Simulation

Before going further, we need to introduce numerical methods which would help us during our simulations. Quantum field theory is a tool used to understand a vast array of perturbative as well as non-perturbative phenomena found in physical systems. Some of the most interesting fields of physics like spontaneous symmetry breaking, studying phase transitions etc, require computation of various observables. Using statistical mechanics, most of these computations will involve large integrals. In many cases, which we will discuss, since exact solutions are not available, we need to resort to perturbative methods using Feynman diagrams. However, in some instances, perturbative methods also fail, the most notable of them being the cases where the coupling constants are not dimensionless, and so in certain regimes, it may happen that the contribution to the integral due to the higher-order terms is comparable to the contribution from the previous term. In those regimes, an alternative computational method is desirable. Monte Carlo methods, using Markov chain based sampling algorithms provide a powerful tool in this regard.

This chapter introduces the computational techniques required to sample data from a probability distribution that is known up to some multiplicative constants. We will see how to apply these algorithms to the integrals in this chapter. To begin, consider a general partition function of a particle in one dimension in some potential $V(x)$ given by

$$Z = \int Dx(t) \exp\{-S_E[x(t)]\}, \tag{2.1}$$

where the Euclidean action for a general potential $V(x)$ is given as

$$S_E[x(t)] = \left(\frac{1}{2}\dot{x}^2 - V(x)\right), \tag{2.2}$$

with periodic boundary conditions (for bosonic fields). Since we are using the canonical ensemble, the system is at temperature given by $T = \beta^{-1}$. Now if we were able to calculate Z, then we can calculate thermodynamic quantities like free energy, internal energy, and other quantities using Z and derivatives of it.

But for a general potential $V(x)$ it cannot be solved exactly. We are all familiar with the numerical methods of integration like the midpoint rule or the Simpson's rule, however, such methods are not feasible for computation of integrals in more than one dimension. Instead of the traditional approach of discretising the integration interval, the Monte Carlo method relies on random numbers.

In this work, we use Monte Carlo Simulations to calculate the Wilson loop operator and implement it using the HMC algorithm. The numerical method that we now know as Monte Carlo was first proposed by Stanislaw Ulam in the 1940s. The Metropolis algorithm that we use was proposed in [Metropolis 53].

Here our main aim is to calculate the expectation values of the observables like Eq (1.6). Since, these observables are usually infinite-dimensional integrals in the continuous case, so to evaluate these, we put the system on a lattice and reduce the integral to a summation in the limit that the lattice size is large enough. Then for an observable O the quantity becomes

$$\langle \mathscr{O} \rangle \approx \langle \mathscr{O} \rangle_n = \frac{\sum_{s=1}^{n} \mathscr{O}(\{x_t\}_s) e^{-S_{lat}(\{x_t\}_s)}}{\sum_{s=1}^{n} e^{-S_{lat}(\{x_t\}_s)}}. \tag{2.3}$$

As a consequence of statistical mechanics, here we can use the probability distribution given by the partition function of the system. Now, we can multiply and divide the observables at each point in space $(x_t)_s$ by a weight function p_s

$$\langle \mathscr{O} \rangle = \frac{\sum_{s=1}^{n} \frac{\mathscr{O}(\{x_t\}_s)}{p_s} p_s e^{-S_{lat}(\{x_t\}_s)}}{\sum_{s=1}^{n} \frac{e^{-S_{lat}(\{x_t\}_s)}}{p_s} p_s}. \tag{2.4}$$

Thus, we can see that each microstate s is generated with the probability p_s and thus, we can approximate $\langle \mathscr{O} \rangle$ with $\langle \mathscr{O} \rangle_n$

$$\langle \mathscr{O} \rangle_n = \frac{\sum_{s=1}^{n} \frac{\mathscr{O}(\{x_t\}_s)}{p_s} e^{-S_{lat}(\{x_t\}_s)}}{\sum_{s=1}^{n} \frac{e^{-S_{lat}(\{x_t\}_s)}}{p_s}}. \tag{2.5}$$

If we look at Eq. (2.4), we will see that $\langle \mathscr{O} \rangle$ will be dominated by terms where the exponential term is large as e^{-x} decays very fast. So it is best to choose the probability distribution such that $p_s \propto e^{-S_{lat}(x_{ts})}$. Its normalised form is given by

$$p_s = \frac{e^{-S_{lat}(\{x_t\}_s)}}{\sum_{s=1}^{n} e^{-S_{lat}(\{x_t\}_s)}}. \tag{2.6}$$

On substituting this in Eq. (2.5), we obtain

$$\langle \mathscr{O} \rangle_n = \frac{1}{n} \sum_{s=1}^{n} \mathscr{O}(\{x_t\}_s) \tag{2.7}$$

which is the equation that we will use in our simulations.

2.1 Naive Sampling

Consider a definite integral $f(x)$ whose integral is given by

$$I = \int_{x_i}^{x_f} f(x)dx. \tag{2.8}$$

This can be found by choosing random points in the configuration space from a uniform probability distribution and the acceptance of the chosen point x would depend on the value of $f(x)$ at that point. They are usually generated using a pseudo-random number generator. The higher the functional value at that point, the more probability of it being by the algorithm. Thus the idea behind this naive sampling method is that since the contribution of the integral is largely due to points with high functional value, therefore the integral can be reasonably approximated using only those points.

In order to find this integral, we need to find a probability distribution, such that it assigns weight to various parts of the configuration space linearly proportional to the part's contribution to the integral. If all the contribution to the integral comes from a fixed point in configuration space, then a Kronecker delta function could be used to find the integral. The idea is that if we visit various parts of the configuration space during a random walk, we would be more likely to visit the places with a higher probability in the distribution, so ideally, the integral can be approximated more accurately using that probability distribution. However, the challenging part is to find such a probability distribution, if at all, it exists.

Monte Carlo method is preferred to other numerical integration methods like the Simpson's rule when integration is over d-dimensional space as for. The error in Monte Carlo methods goes down at a rate in the order of $O(\frac{1}{\sqrt{N}})$ where N is the total no of points. Thus, we see that the larger the no of points, the more is the method useful for us. So, if we want to exactly approximate an integral in the continuum limit, Monte Carlo method is more computationally efficient and accurate. Monte Carlo methods are heavily dependent on the probability distribution used. For instance, if the probability distribution used to approximate the integral is a Cauchy distribution

$$f(x) = \frac{1}{\pi(1+x^2)}, \tag{2.9}$$

which is frequently seen in spectroscopy and looks like a normal distribution, but it has no defined mean or variance. So, we cannot evaluate such an integral because error analysis is impossible. Also, since a probability distribution is always normalised, it is very difficult to approximate integrals that cannot be normalised, for instance, integrands that oscillate periodically.

2.2 Importance Sampling

In this section we discuss a method that can increase the efficiency of Monte Carlo integration. This technique is called importance sampling. It is one of the several available variance reduction techniques, in the context of Monte Carlo integration. In order to increase the efficiency of Monte Carlo methods, it is useful if we can make the function more flat (non-oscillating). In naive sampling, we used random points in the configuration space where each point is treated equally. However, in importance sampling we use a weight function $w(x)$ that assigns weight to various points in the configuration space based on their importance in calculating the integral [Joseph 19]

$$\int_{x_i}^{x_f} w(x) = 1. \tag{2.10}$$

Thus the random points x_r are selected according to the probability density. The integral is given by

$$\int_{x_i}^{x_f} w(x)f(x)dx \approx \frac{1}{N}\sum_{r=1} Nf(x_r). \tag{2.11}$$

An important drawback of this sampling method is that it is not possible to give any theoretical bound on the truncation error. It would entirely depend on how "well" the probability distribution approximates the integral and to calculate that, we need to know the integral and if we knew the integral value, there would be no reason to use Monte Carlo method! We can however measure the probable error given by the Central Limit Theorem. For the integral I, the variance σ^2 is defined as

$$\sigma^2 = \int_{x_i}^{x_f} (f(x)-I)^2 w(x)dx = \int_{x_i}^{x_f} [f^2(x)w(x)-I^2]dx. \tag{2.12}$$

This comes out to be of the order of $O(\frac{1}{\sqrt{N}})$.

Let us consider integral of a well-behaved function $f(x)$

$$I = \int_0^1 f(x)dx \tag{2.13}$$

and rewrite it as

$$I = \int_0^1 dx \left[\frac{f(x)}{px}\right] p(x). \tag{2.14}$$

Here $p(x)$ is a weight function and we can approximate the integral by

$$I \approx \frac{1}{N}\sum_{r=1}^{N} \frac{f(x_r)}{p(x_r)}. \tag{2.15}$$

Here we can use the normalised version of $f(x)$ as the weight function, but since that is not possible, as an alternative, we could also select $p(x)$ as some approximation to the above function. Even after doing that we will end up with a small variance. The main problem with importance sampling method is the difficulty of generating random numbers from a probability density function $p(x)$, particularly in several dimensions. If we use a $p(x)$ whose integral is known and whose behaviour approximates that of $f(x)$, then we can expect to reduce the variance. Thus, importance sampling is choosing a good distribution from which to simulate our random variables. It involves multiplying the integrand by 1 to give an expectation of a quantity that varies less than the original integrand, over the region of integration.

A good importance sampling function $p(x)$ should have the following properties:

- $p(x) > 0$ whenever $f(x) \neq 0$
- $p(x)$ should be close to being proportional to $|f(x)|$
- it should be easy to draw values from $p(x)$
- it should be easy to compute the density $p(x)$ for any value x that we might realise.

As the dimensionality of the integral increases, where Monte Carlo techniques are most useful, fulfilling the above set of criteria can turn out to be quite a non-trivial endeavour.

Another problem that crops up is when the tail of $p(x)$ slows down much faster than $f(x)$. From Eq. (2.15) above, we can see that it will blow up in those regions, and consequently, they will receive importance while estimating the integral. So, this method is not always reliable. A much more efficient way is the Markov Chain Monte Carlo (MCMC) which we will elaborate now.

2.3 Markov Chain Monte Carlo

Unlike naive and importance sampling methods in Monte Carlo, the goal of MCMC is not to sample a region uniformly, but rather to visit a point x with a probability proportional to some distribution $\pi(x)$. This distribution is not a probability in the sense that it is not normalised, however, it is proportional to a probability that puts its sample points preferentially at points where $\pi(x)$ is large. Thus it greatly reduces the computational time in multi-dimensional space. In MCMC, we sample distribution using a Markov chain. A Markov chain is a sequence of points, $x_0, x_1, x_2, \ldots$, that are locally correlated, with the ergodic property [Robert 04].

The sequence of points will eventually visit every point x, in proportion to $\pi(x)$. Here Markov means that each point x_i is chosen from a distribution that depends only on the value of the immediately preceding point x_{i-1}. Thus the chain is completely defined by a transition probability function between two variables, $p(x_i|x_{i-1})$.

Transition rate is defined as a product of the population density and a transition probability.

If $p(x_i|x_{i-1})$ is chosen to satisfy the detailed balance equation

$$\pi(x_1)p(x_2|x_1) = \pi(x_2)p(x_1|x_2), \tag{2.16}$$

where there is no net transition, then the Markov chain will in fact sample $\pi(x)$ ergodically.

Integrating both sides with respect to x_1 gives

$$\int dx_1 \pi(x_1)p(x_2|x_1) = \pi(x_2)\int p(x_1|x_2)dx_1 = \pi(x_2). \tag{2.17}$$

The left-hand side of the above equation is the probability of x_2, computed by integrating over all possible values of x_1 with the corresponding transition probability. The right-hand side is the desired $\pi(x_2)$. Thus Eq. (2.17) says that if x_1 is drawn from π, then so is its successor x_2, in the Markov chain.

Let us consider a Markov chain consisting of a sequence of random elements, $x_0, x_1, x_2, \ldots$, of some set. As noted before, the chain has the property that the conditional distribution of x_i, given $x_0, x_1, x_2, \ldots, x_{i-1}$ depends only on x_{i-1}. The set in which $x_0, x_1, x_2, \ldots$ take their values is called the state space S of the Markov chain. A state space could be finite or infinite. Markov chains exhibit the so-called Markov property or memoryless property. We are interested in finding a stationary distribution, $\pi(x)$, starting from an initial distribution say, $\mu(x_0)$. This is done through a series of Markov chain steps, jumping from x_i to x_{i+1}. This is done with a transition probability function p. It is a matrix in the discrete case and a kernel in the continuous case. For a Markov chain to have a stationary distribution, it must be irreducible (each state must be accessible from any other state in finite time) and aperiodic (if in any Markov chain sequence, no state repeats itself).

If both irreducibility and aperiodicity are respected, by a finite-state Markov chain, then there exists a stationary distribution.

Theorem 1 (Uniqueness Theorem) *Any irreducible and aperiodic Markov chain has exactly one stationary distribution.*

Theorem 2 (Fundamental Limit Theorem) *Let $x_0, x_1, \ldots$ be an irreducible and aperiodic Markov chain, with state space*

$$S = s_1, s_2, \ldots, s_k$$

and a transition matrix (or Markov matrix) P. It also has an arbitrary initial distribution $\mu(0)$. Then, for any distribution π, which is stationary distribution for P, we have

$$\mu(n) \to \pi$$

with $n \geq 0$.

These two theorems guarantee that if we start from an element x_0 drawn from an initial distribution $\mu(x)$. Then the distribution of x_n will converge to $\pi(x)$ in finite time and we can say that the system has reached an equilibrium.

A Markov chain with stationary distribution π is called reversible if its transition matrix (or kernel) P is such that it exhibits detailed balance. It means that the chain would look the same if we ran it forwards or backward in time. Reversibility is a property of a distribution on the state space S, which is related to the transition matrix P. It is important to note that stationarity does not imply reversibility. The transition matrix represents the Markov chain as a matrix containing the transition probabilities. The matrix is normalised such that the elements of any given column add up to 1. Thus, these theorems guarantee that MCMC algorithms always converge. Thus, we have to find an efficient MCMC algorithm that leads to the convergence of the Markov Chain efficiently. There are several algorithms available like the Gibbs algorithm etc, but we will be using the Hamiltonian Monte Carlo.

2.3.1 Hamiltonian Monte Carlo

Metropolis algorithm is one of the algorithms to sample data from probability distributions which are known up to some multiplicative constant [Metropolis 53]. Metropolis algorithm uses Markov chain to generate these configurations and in the limit of Markov chain length going to infinity, it converges to the exact probability distribution. However, this algorithm uses the principle of random walks for sampling the distribution, which is not computationally efficient for computing large integrals. So we use a modified version of the Metropolis algorithm, the Hamiltonian Monte Carlo.

This algorithm was first used to study molecular dynamics. In stead of analysing random paths as in the original Metropolis algorithm, this algorithm makes use of directed paths, which are given by the Hamilton's equations of a system. In order to formulate Hamilton's equations, we need the Hamiltonian of a system, and for that, we need to introduce conjugate momentum variables for each field variable and a fictitious time τ to evolve the field and conjugate variables using Hamilton's equations. Let us say that the probability distribution we want to sample from is given by $P(\vec{x} = \{x_1, x_2, \ldots, x_n\})$, which can be written as $e^{-\{-\log P(\vec{x})\}}$. Thus if for each variable x_i, we introduce the conjugate momentum variable p_i, we get the Hamiltonian as

$$H(x,p) = \frac{1}{2}\sum p_i^2 - \log P(\vec{x}). \tag{2.18}$$

The joint probability distribution $\pi(x,p)$ is given by the exponential of the Hamiltonian and in order to recover $P(\vec{x}$, we need to integrate out the momentum variables p_i. Hamilton's equations are given as

$$\frac{d\vec{x}}{d\tau} = \frac{\partial H}{\partial \vec{p}}, \tag{2.19}$$

$$\frac{d\vec{p}}{d\tau} = -\frac{\partial H}{\partial \vec{x}}. \tag{2.20}$$

These equations are used to evolve the field variables and the conjugate momentum variables in time τ numerically. The integrators to evolve the Hamiltonian must be symplectic, i.e., they preserve the time reversibility and the volume of phase space, which are characteristic properties of the Hamiltonian. The simplest integrator of this form is the *leapfrog* integrator. These are the steps followed in every leapfrog iteration

$$p_i(\tau + \frac{\varepsilon}{2}) = p_i(\tau) - \frac{\varepsilon}{2}\frac{\partial H(\vec{q}(\tau))}{q_i}, \tag{2.21}$$

$$q_i(\tau + \varepsilon) = q_i(\tau) + \varepsilon p_i(\tau + \frac{\varepsilon}{2}), \tag{2.22}$$

$$p_i(\tau + \varepsilon) = p_i(\tau + \frac{\varepsilon}{2}) - \frac{\varepsilon}{2}\frac{\partial H(\vec{q}(\tau + \varepsilon))}{q_i}. \tag{2.23}$$

Suppose the probability distribution we want to sample from is $P(\vec{x} = \{x_1, x_2, \ldots, x_n\})$, which depends on n variables. Then the steps of the HMC algorithm is given as

1. First, we construct the Hamiltonian as given by Eq. (2.18). Then we choose some initial state $\vec{x_0} = \{x_1^0, x_2^0, \ldots, x_n^0\})$.

2. Then, we choose $p_i = p_i(0)$ randomly from a standard Gaussian distribution. Then we compute $H_i(0) = H(\vec{x_0}, \vec{p_0})$.

3. Now we evolve the $\vec{x_i}$ and $\vec{p}$ using leapfrog algorithm for n time steps using ε as the time resolution parameter. The $\vec{x_0}(n\varepsilon)$ is the new proposed state. Then we calculate $H_f = H(\vec{x_0}(n\varepsilon), \vec{p}(n\varepsilon)$.

4. Now we apply the Metropolis step. $A(\vec{x_i}(n\varepsilon), \vec{x_i}(0) = \min\left\{1, \frac{\pi(\vec{x_0}(n\varepsilon), \vec{p}(n\varepsilon)}{\pi(\vec{x_0}(0), \vec{p}(0)}\right\} = e^{-\{H_f - H_i\}}$. We accept the $\vec{x_i}(n\varepsilon)$ state with probability $A(\vec{x_i}(n\varepsilon), \vec{x_i}(0)$. If the state is accepted, we have $\vec{x_{i+1}}(0) = \vec{x_i}(n\varepsilon)$, else $\vec{x_{i+1}}(0) = \vec{x_i}(0)$.

5. Repeat steps 2-4.

Since the Hamiltonian in Eq. (2.18) is time-reversible i.e., $H(p,q) = H(-p,q)$, this ensures that the detailed balance condition is obeyed. Since the Hamiltonian is also symplectic, we need to ensure that the volume of phase space remains preserved at every iteration which is necessary to ensure that the joint probability distribution $\pi(\vec{q}, \vec{p})$ remains covariant under the Hamiltonian dynamics, since the Jacobian of the transformation is 1.

For implementing the Metropolis step, we need to use a uniform random number generator. We use the `ran3` function from the text *Numerical Recipes* for this purpose. We choose a random number $\alpha \in (0,1)$. If $\alpha \leq A(\vec{x_{prop}}, \vec{x_i})$ then the proposed state is accepted, else the proposed state is rejected. Detailed balance condition is obeyed in the Metropolis test. i.e., $P(\vec{x_1})Q(\vec{x_2}|\vec{x_1})A(\vec{x_2},\vec{x_1}) = P(\vec{x_2})Q(\vec{x_1}|\vec{x_2})A(\vec{x_1},\vec{x_2})$

$$\text{If} P(\vec{x_2}) > P(\vec{x_1}) \implies A(\vec{x_2},\vec{x_1}) = 1 \text{ and } A(\vec{x_2},\vec{x_1}) = \frac{P(\vec{x_1})}{P(\vec{x_2})Q(\vec{x_2}|\vec{x_1})A(\vec{x_2},\vec{x_1})} \tag{2.24}$$

$$\implies P(\vec{x_1})A(\vec{x_2},\vec{x_1}) = P(\vec{x_1}) \tag{2.25}$$

$$P(\vec{x_2})A(\vec{x_1},\vec{x_2}) = P(\vec{x_2})\frac{P(\vec{x_1})}{P(\vec{x_2})} = P(\vec{x_1}) \tag{2.26}$$

$$\text{So} P(\vec{x_1})Q(\vec{x_2}|\vec{x_1})A(\vec{x_2},\vec{x_1}) = P(\vec{x_2})Q(\vec{x_1}|\vec{x_2})A(\vec{x_1},\vec{x_2}) \tag{2.27}$$

Since Monte Carlo method is based on random sampling, it is useful to have measures to ascertain the reliability of the data obtained. One of the tests to check if the HMC is implemented properly is to look for the expectation value of $e^{-\Delta H}$. The value is given by [Ydri 17]

$$\left\langle e^{-\Delta H}\right\rangle = \left\langle e^{-(H(\vec{x}',\vec{p}')-H(\vec{x},\vec{p}))}\right\rangle \begin{aligned} &= \frac{1}{Z}\int d\vec{p}d\vec{x}e^{-H(\vec{x},\vec{p})}e^{-(H(\vec{x}',\vec{p}')-H(\vec{x},\vec{p}))} \\ &= \frac{1}{Z}\int d\vec{p}d\vec{x}e^{-H(\vec{x}',\vec{p}')} \\ &= \frac{1}{Z}\int d\vec{p}'d\vec{x}'e^{-H(\vec{x}',\vec{p}')} \end{aligned} \tag{2.28}$$

$$\left\langle e^{-\Delta H}\right\rangle = \frac{1}{Z}Z = 1. \tag{2.29}$$

Thus, if the expectation value of $e^{-\Delta H}$ fluctuates about 1, then HMC algorithm has been implemented correctly.

After thermalising, we can use the field configurations to calculate the expectation values of observables using Eq. (2.7). But the error in the expectation value depends on whether the states used to calculate the expectation are correlated or not. This is because in uncorrelated states $\langle x_i x_j\rangle \neq \langle x_i\rangle\langle x_j\rangle$, hence the terms do not cancel while evaluating the variance of such a data set. If the states $\{x_1, x_2, \ldots, x_N\}$ are not correlated, then error in the expectation of f is given by [Gattringer 07]

$$\delta f = \frac{\sigma}{\sqrt{N}}, \tag{2.30}$$

where σ is the standard deviation in the expectation value of f. We can get the uncorrelated states by finding the autocorrelation length and taking the states for expectation values after the autocorrelation length. Autocorrelation length is where autocorrelation becomes nearly zero.

2.4 Simulation Details

For all Monte Carlo simulations, certain features remain common. We will discuss some of these basic features here and additional features will be discussed in the later chapters.

2.4.1 Thermalisation and Initial Condition

Theorems 1 and 2 guarantee that we can initialise our simulation with any field configuration for both the algorithms, and we will get the same result from any starting point as the algorithm respects the ergodic property. But sometimes the algorithm may take a longer time to converge to the desired probability distribution for some starting points. So, if we have some intuition regarding the probability distribution then we can use that for selecting the initial condition. But mostly we do not have earlier information so we mostly make a cold start for the simulation. A cold start is an initial configuration in which all the field configurations are set to zero (for bosonic fields only). If we start with a random field configuration, then we call it a hot start. Sometimes people use cold start for some fields and hot start for some fields, this is known as mixed-field configuration. In all our simulations, we have used the cold start, since no information was available to us about the nature of the final solution. After choosing the initial field configurations, we run the simulation to allow the distribution to reach the equilibrium distribution and after that, we use it to generate/sample data. This process is known as thermalisation. Thermalisation is achieved when the value of any measured observable starts to oscillate about some value. Ideally, such a measured observable should have a contribution from all the field variables. Then we can use the data obtained after thermalisation to calculate the expectation values. Thermalisation occurs faster in HMC as compared to the Metropolis algorithm since random walk Markov chains take longer to converge as compared to directed Markov chains.

2.4.2 Sweeps and Updating

On the lattice, we choose to update only one field variable at one lattice site at a time and repeat this for other lattice sites. For the Metropolis algorithm it is better to update the field variable at one lattice site multiple times before moving to the next field variable. This is done so that the system converges to the required distribution faster. For HMC it does not give any better results by making multiple changes at one lattice site. So we update the field variables only once at each lattice site for HMC. Thus in the leapfrog algorithm, we set n to 1 When we have updated the whole lattice with the new configuration it is known as a sweep. We collect all the information of the observables after each sweep and store this data in a file. Every thermalisation plot is plotted against the sweep number. To prevent any data loss we store the field configurations after every 200 sweeps. This way we did not need to thermalise the system again and again and we can restart the simulation with this previously stored configuration.

2.4.3 Autocorrelation

We have to take into consideration the fact that field configurations generated by the Markov process are not independent since each configuration is dependent on the previous configuration. This is because we are using pseudo-random numbers in the Metropolis test, which has a periodic nature, so it is expected that the values generated would be correlated. So we cannot directly evaluate our observable from these correlated field configurations as this will increase error bars and it would require more field configurations to reduce these error bars. In order to mitigate this error, we run the simulation for sufficient number of times and store the values of observables for every sweep. Since, in a Markov chain in equilibrium, the autocorrelation solely depends on the time step [Gattringer 07], we calculate the autocorrelation and the normalised autocorrelation to get to know how many sweeps are required to generate a statistically independent field configuration. Autocorrelation and normalised autocorrelation to the first leading-order correction are defined as

$$\Gamma_\tau = \frac{1}{N-\tau}\sum_{i=1}^{N-\tau}(f_i - \langle f \rangle)(f_{i+1} - \langle f \rangle) \tag{2.31}$$

$$\rho_\tau = \frac{\Gamma_\tau}{\Gamma_0}, \tag{2.32}$$

where N is the number of data points after the thermalisation, τ is the number of sweeps for which we are checking the correlation. Σ_0 is simply the variance. Autocorrelation length is defined as the minimum number of sweeps required to get the uncorrelated field configuration. So we find the autocorrelation length by looking at the values of the normalised autocorrelation where it crosses 0 or just near 0. After getting the autocorrelation length we calculate the expectation value of the observable and error bars from these uncorrelated field configurations. From now on we will use the term autocorrelation for the normalised autocorrelation.

Chapter 3

Equivalence of Wilson Loop with Schwarzschild Radius

Superstring theory was first formulated to explain all of the particles and fundamental forces of nature in one theory by modelling the forces as well as the various fundamental particles as vibrations of tiny supersymmetric strings. Superstring theory is short for supersymmetric string theory and it is named so because it takes into account both fermions and bosons and incorporates supersymmetry to model gravity. The symmetry known as supersymmetry occurs in such theories because there is a corresponding fermionic superpartner for every boson and vice-versa, and as a consequence, the action is invariant under two super-symmetry charges, Q and $\bar{Q}$. That is, $QS = \bar{Q}S = 0$.

In string theory, we frequently come across worldsheets, which are two-dimensional manifolds that describe the trajectories of a string in spacetime. D-branes are physical hyper-planes where open strings end and can warp space as they possess mass and charge. Supergravity, which is not valid at short distances, can describe the long-range fields of D-branes as long as the curvatures are locally small compared to the Planck length.

3.1 Correspondence between String Theories and Matrix Quantum Mechanics

Type II string theory accounts for two of the five consistent superstring theories in ten dimensions. Both the theories have the maximal amount of supersymmetry in ten dimensions and are based on oriented closed strings i.e., they form directed loops. The Type IIA and Type IIB theories differ only in choice of GSO projection. GSO projection is a selection of possible vertex operators which projects out those operators which have the properties of closure, mutual locality and modular invariance.

As explained in [Banks 97], the uncompactified eleven-dimensional M-theory and the supersymmetric matrix quantum mechanics at the large N limit describing $D0$ branes are equivalent. The evidence for the conjecture manifests itself as several correspondences between the two theories. One of the most important correspondence worth noting is that the compactification radius R is related to the string coupling constant by

$$R = g^{\frac{2}{3}} l_p = g l_s, \tag{3.1}$$

where l_s is the string length scale

$$l_s = g^{-\frac{1}{3}} l_p. \tag{3.2}$$

This is one of the most crucial properties that makes the $D0$-branes an ideal candidate describing M-theory in the infinite-momentum limit. As a consequence of supersymmetry, the simple matrix model is sufficient to describe the properties of the entire Fock space sparsely populated by massless particles of the supergravity theory. In the region of infinite coupling, M theory changes into an eleven dimensional theory in a background of infinite flat space.

3.2 Wilson Loop

Wilson loop is defined for a closed loop in spacetime. In numerical form, it is the trace of the product of two Wilson lines and two temporal transporters [Gattringer 07]. It can be represented as the path-ordered exponential of the gauge field

$$W = \frac{1}{N} \mathrm{Tr} \mathscr{P} \exp\left(\imath \oint A_\mu dx_\mu \right), \tag{3.3}$$

with the trace in the fundamental representation. Since the Wilson loop can be defined for any closed path in space one can obtain a large class of gauge-invariant operators. These operators, and their products, form a complete basis of gauge-invariant operators for pure Yang-Mills theory [Drukker 99].

The Wilson lines connects two spatial points along a path with all link variables restricted to a temporal point, while the temporal transporters are the line of all link variables restricted to a spatial position. The trace ensures that the observable is gauge-invariant, which is helpful in reducing computational time for evaluating an integral (else we would have to integrate over the gauge group) unless we fix the gauge. If the Wilson loop is restricted along only one spatial direction, it is called planar. It is shown in [Gattringer 07], that the Wilson lines are the quark/anti-quark propagators in the limit of infinitely-massive quarks.

Since it is gauge-invariant there are several applications of the Wilson loop. We have previously referred to the fact that an infinitely massive quark in the fundamental representation moving along the Wilson loop will be transformed by the phase factor W. This implies that the dynamical effects of the gauge dynamics on external quark sources can be measured by the Wilson loop. For a parallel quark anti-quark pair, the Wilson loop is the exponent of the effective potential between the quarks and serves as an order parameter for confinement.

3.3 Construction of Wilson Loop Operator

To construct the Wilson loop, we separate a single D-brane from the bunch of N D-branes and keep them at a very large distance. For large N, we can ignore the fields on the distant D-brane, except for open strings stretching between it and the other N D-branes. The ground states of these open strings are the W bosons and their superpartners of the broken, $SU(N)$, gauge group [Drukker 99]. Therefore, the trajectories of these strings should be understood in the same way as that of an infinitely massive particle in the fundamental representation.

Here, the Wilson loop is derived by calculating the correlation function of the matter fields in terms of the path integral over the particle's trajectory. The resulting phase factor determines how the Wilson loop is coupled to the gauge field. However, we have already noted that the $\mathscr{N}=4$ super Yang-Mills theory in 4 dimensions does not contain such fields. Instead, we use W bosons which are massive and appear once we break $SU(N+1) \to SU(N)\times U(1)$.

According to the Maldacena conjecture [Maldacena 98], the expectation value of the Wilson loop operator can be written in terms of the string action along the boundary of the loop. This is

$$\langle W[\mathscr{C}]\rangle = \int_{\partial X=\mathscr{C}} DX \exp\Big(-\sqrt{\lambda}S[X]\Big), \tag{3.4}$$

for some string action $S[X]$. Here X represents both the bosonic and the fermionic coordinates of the string. For large λ, we can estimate the path integral by the steepest descent method. Consequently, the expectation value of the Wilson loop is related to the area A of the minimal surface bounded by $\mathscr{C}$ as

$$\langle W\rangle \approx \exp\Big(-\sqrt{\lambda}A\Big). \tag{3.5}$$

The motivation for this ansatz is that the W boson considered here can be described in the D-brane language as an open string going between the single separated D-brane and the other N D-branes. Near the event horizon, the N D-branes are replaced by the AdS_5 geometry and the open string is stretched from the boundary to the interior of AdS_5. The Wilson Loop obeys the constraint $\dot{x}^2 = \dot{y}^2$, which is related to the coupling of the fundamental string to the gauge and the scalar fields [Maldacena 98].

Thus we come to the conclusion that the ground states of the string stretched from the distant D-brane to the others consist of the W-bosons and their superpartners in the fundamental representation of the gauge group of the remaining branes. In the classical limit, the string is described by a minimal surface [Drukker 99]. However, the curvature of AdS_5 ensures that the minimal surface does not stay near the boundary, but goes deep into the interior of space, where the area element can be arbitrarily reduced. Consequently, the behaviour of the Wilson loop, for large area, is that of a conformal theory, and the area law does not produce confinement. Thus, for large λ, the expectation value of the Wilson loop is related to the classical action of the string, with appropriate boundary conditions. To the leading order in λ, we can ignore the effect of the other fields (Ramond-Ramond) and use the Nambu-Goto action, or in other words, the area of the minimal surface

$$A = \int \frac{d\sigma_1 d\sigma_2}{2\pi\alpha'\sqrt{\lambda}}\sqrt{g} = \int \frac{d\sigma_1 d\sigma_2}{2\pi Y^2}\sqrt{\det(\partial_\alpha X^\mu \partial_\beta X^\mu + \partial_\alpha Y^i \partial_\beta Y^i)}. \tag{3.6}$$

Because of the Y^{-2} factor in the integral, this area is infinite. After regularising the divergence, the infinite part was identified as due to the mass of the W boson and subtracted [Maldacena 98]. If one considers the two parallel lines (with opposite orientation) as a quark-anti quark pair, the remaining finite part defines the quark-anti quark potential. Such calculations were used to study the phases of the $\mathcal{N} = 4$ super Yang-Mills theory and to demonstrate confinement in non-supersymmetric generalisations. This is roughly the procedure that is followed and the requisite calculations are shown below.

The procedure here is similar to the one described in [Nishimura 07]. As explained earlier, the gauge theory in the fundamental representation is free of any particles. Since there is $SO(N)$ symmetry in our system, for simplicity, we can assume that the Wilson loop lies along one direction in spacetime [Pestun 07].

For general D-branes the N D-branes, which are placed on top of each other creates a curved background geometry since D-branes are massive objects. We consider a string stretched between the probe D-brane and one of the N D-branes. We should note here that D-branes are the charged objects that also act as sources for Ramond-Ramond fields [Polchinski 98], however we ignore them during our calculations.

The amplitude for the string propagating along a certain loop $\mathscr{C}$ on the D-brane can be calculated in two different ways. According to the world-volume theory of the N D-branes the process is viewed as a heavy test particle in the fundamental representation of the $U(N)$ group propagating along the loop and the amplitude is given by

$$\mathscr{A} = \langle W(\mathscr{C})\rangle\, e^{-M\mathscr{L}}, \tag{3.7}$$

where $W(\mathscr{C})$ represents the Wilson loop associated with the loop $\mathscr{C}$, whose perimeter has the length $\mathscr{L}$. The mass M of the test particle is given by the distance of the N D-branes and the probe D-brane.

On the gravity side, we can view the process as a string propagating along a curved spacetime background. The amplitude is calculated as

$$\mathscr{A} = \frac{1}{N}\int_{\mathscr{C}} e^{-S_{string}}, \tag{3.8}$$

where S_{string} represents the worldsheet action whose bosonic part is given by the Polyakov action [Zwiebach 04]

$$S_P = \frac{1}{2\pi\alpha'}\frac{1}{2}\int d^2\sigma\sqrt{h}(h^{ab}g_MN\partial_a x^M\partial_b x^N + \alpha'\phi R_{(2)}). \tag{3.9}$$

Here x^M represents the embedding of the worldsheet into the target space with the metric $g^{MN}(M,N = 1,\dots 10)$, while $R_{(2)}$ represents the two-dimensional scalar curvature defined for the worldsheet metric $h^{ab}(a,b = 1,2)$. The effective string coupling is given as e^{ϕ} in terms of the dilation field ϕ.

In the following treatment, we concentrate most in the parameter region in which the string coupling is so small that we only have to consider the disk amplitude in 3.8. We also restrict ourselves to small α', which corresponds to large string tension, so that the path integral Eq. (3.8) is dominated by the saddle-point configuration, i.e., the strings stretched between the D-branes is free from any kinks. In this parameter region, one can use the classical solution to the supergravity as the background. If we take into account the string/gauge duality, the parameter region corresponds to taking the planar large-N limit with large 't Hooft coupling constant on the gauge theory side.

Equating Eq. (3.7) and (3.8), we obtain a formula which relates the Wilson loop in the strongly coupled gauge theory to the string amplitude on the classical background geometry. The obtained result indeed agrees with the prediction from the gravity side.

Now, we concentrate solely upon the $D0$-brane case, and we consider the loop $\mathscr{C}$ to be winding once around the temporal direction. The corresponding Wilson loop operator in the gauge theory is given by

$$W = \frac{1}{N}\mathrm{Tr}\left(\mathrm{P}e^{\int_0^{\beta}(\imath A(t)+n_i X_9(t))}\right), \tag{3.10}$$

where n is a unit vector in R_9 specifying the direction in which the probe $D0$-brane is separated and P is the path-ordering operator. Here, to simplify our calculations, we can consider the Wilson loop to be lying along the x_9 direction, i.e., $n_i = \delta_{i9}$. Here, the adjoint scalar appears unlike the definition of the Polyakov line since the end of the string is coupled not only to the gauge field but also to the adjoint scalar. The overall factor $\frac{1}{N}$ is introduced to make the quantity finite in the planar large-N limit.

In general relativity, a black brane is a solution of the equations that generalises a black hole solution but it is also extended – and translationally symmetric – in p additional spatial dimensions. That type of solution would be called a black p-brane. In string theory, the term black brane describes a group of $D1$-branes that are surrounded by a horizon. With the notion of a horizon in mind as well as identifying points as 0-branes, a generalisation of a black hole is a black p-brane.

The metric for the gravity dual of the supersymmetric Matrix Quantum Mechanics (MQM) is given by the near-horizon geometry of the (Euclidean) near-extremal black 0-brane solution in type IIA supergravity

$$\frac{ds^2}{\alpha'} = \frac{U^{\frac{7}{2}} f(U)}{\sqrt{d_0 \lambda}} dt^2 + \frac{\sqrt{d_0 \lambda}}{U^{\frac{7}{2}} f(U)} dU^2 + \frac{\sqrt{d_0 \lambda}}{U^{\frac{3}{2}}} d\Omega_8^2 \tag{3.11}$$

where $f(U) = 1 - \frac{U_0^7}{U^7}$ and $d_0 \equiv 2^7 \pi^{\frac{9}{2}} \Gamma(7/2)$. The Schwarzschild radius R_{Sch} and the inverse Hawking temperature β are given by

$$R_{Sch} = \alpha' U_0 \tag{3.12}$$

$$\beta = \frac{4}{7} \pi \sqrt{d_0 \lambda} U_0^{\frac{5}{2}} \tag{3.13}$$

respectively. The string disk-amplitude in the background geometry is evaluated after we consider a string worldsheet localised in the S_8 direction.

In the $\alpha' \to 0$ limit, and replacing the string action by the Nambu-Goto action (area of the string worldsheet times the string tension), we get the identity

$$\log \langle W(\mathscr{C}) \rangle - \beta M = \frac{\beta U_0}{2\pi} - \frac{\beta U_\infty}{2\pi}, \tag{3.14}$$

where U_∞ represents the position of the probe $D0$-brane. It is natural to identify the second terms on both sides [Drukker 99]. Thus we obtain

$$\log \langle W(\mathscr{C}) \rangle = \frac{\beta U_0}{2\pi} = \frac{\beta R_{Sch}}{2\pi \alpha'} = 1.89 \left(\frac{T}{\lambda^{\frac{1}{3}}} \right)^{-\frac{3}{5}}. \tag{3.15}$$

3.4 Error and Range of Validity

To find the range in which these equation is valid we start from the range of validity in the supergravity description [Nishimura 07]. By changing the target-space coordinates as $U = U_0 u^{\frac{2}{5}}$ and that the geometry is asymptotic at large u to a geometry which is conformally equivalent to $AdS_2 \times S^8$ [Jevicki 99], and that the typical length scale of the geometry is given by $\rho \equiv (\frac{uT}{\lambda^{\frac{1}{3}}})^{-\frac{3}{10}} \alpha'$. This scale should be much larger than the string length $\alpha'^{\frac{1}{2}}$ for the α' corrections to the supergravity action to be negligible. Hence, we obtain $\frac{uT}{\lambda^{\frac{1}{3}}} \ll 1$. In this case, the first term in Eq. (3.9), becomes large, and the semi-classical treatment for the string amplitude (3.8) is also justified. Note, however, that we have introduced U_∞ as the coordinate of the probe $D0$-brane. Assuming that we only need to require $\frac{U_\infty}{U_0}$ to be large (but finite), we may assume u to be finite as well. Then we obtain the condition

$$\frac{T}{\lambda^{\frac{1}{3}}} \ll 1. \tag{3.16}$$

To ensure that the effective string coupling is small, from (3.16), we obtain $N^{-\frac{10}{21}} \ll \frac{T}{\lambda^{\frac{1}{3}}}$ noting that $u \geq 1$ in our finite temperature set-up.

We now turn our attention to the possible sub-leading terms in Eq. (3.15) due to corrections on the gravity side. There are three effects one should consider:

1. Coupling with the background field represented by the second term in Eq. (3.9) yields a constant term and a logarithmic term with respect to $\frac{T}{\lambda^{\frac{1}{3}}}$ in Eq. (3.15) as one can see from Eq. (3.9) and Eq. (3.16). The constant term includes $\log N$, but this is cancelled by the pre-factor $\frac{1}{N}$ in Eq. (3.8) as it should.

2. Corrections to the background fields that appear in the action Eq. (3.8). This effect can be neglected at this order since corrections to the type IIA supergravity action starts only at the 3rd order.

3. Quantum fluctuation of the string worldsheet including fermionic degrees of freedom in evaluating Eq. (3.8). This effect yields a constant term to Eq. (3.16).

Thus we obtain a constant error term and thus Eq. (3.15) can be approximated as a straight line.

Chapter 4

Simulations of $D = 4$ and BFSS Matrix Models

Bosonic BFSS is the BFSS matrix model without the fermions. The action of the model is given as

$$S_E = \frac{N}{\lambda} \int_0^\beta dt \ \mathrm{Tr} \left[\frac{1}{2}(D_t X_i)^2 - \frac{1}{4}[X_i, X_j]^2 \right]. \tag{4.1}$$

In order to numerically simulate it on the lattice we need to discretise the action. We reduce the temporal direction to a one-dimensional lattice and that lattice is given by $\Lambda = \{n | n = 0, 1, \ldots, T-1; t = na\}$. Imposing periodic boundary conditions results in identifying 0-th with T-th lattice site and $\beta = aT$. This gives

$$\partial_t X_n^i \to \frac{X_{n+1}^i - X_n^i}{a}. \tag{4.2}$$

To discretise the covariant derivative we introduce link fields U

$$U_{n,n+1} = \mathscr{P} \left\{ exp \left[\imath \int_{na}^{(n+1)a} dt A(t) \right] \right\} = \exp(\imath a A(t)). \tag{4.3}$$

In order to perform numerical simulations, we must discretise time, at the end of which the temporal co-ordinate takes on the form of a periodic lattice. In this section, we try to describe how to implement gauge fields on a lattice by using an example of $D = 4$ Model whose action is obtained by dimensionally reducing the $3+1$-dimensional Yang-Mills model with gauge group $SU(N)$ to $0+1$ dimensions by dimensional reduction described in detail in [Tanwar 20]. The lattice formulation is a necessary prerequisite for using Monte Carlo simulations to analyse such actions which are hard to study otherwise.

4.1 The $D = 4$ Model

First of all, since the algebra of supersymmetry contains continuous translations, which are broken to discrete ones, it seems unavoidable to break it on the lattice. Then, one has to include all the relevant terms allowed by symmetries preserved on the lattice, and fine-tune the coupling constants to arrive at the desired supersymmetric fixed point in the continuum limit. Recent progress is that one can reduce the number of parameters to be fine-tuned (even to zero in some cases) by preserving some part of supersymmetry. In lower dimensions, one can alternatively take the advantage of super-renormalizability, and determine the appropriate counter-terms by perturbative calculations to avoid fine-tuning.

The $D = 4$ model is a toy model and is usually used The Euclidean action of the $D = 4$ model is given by

$$S_E = \frac{N}{\lambda}\int_0^\beta dt \;\; \mathrm{Tr}\left[\frac{1}{2}(D_t X_i)^2 - \frac{1}{4}[X_i, X_j]^2\right], \tag{4.4}$$

where $D_t X^i = \partial_t X^i - \imath[A(t), X^i]$ is the covariant derivative. As always repeated indices are summed over, $i, j = 1, 2, \ldots, d$, $\lambda = Ng^2$, where λ is the 't Hooft coupling and g is the one-dimensional Yang-Mills coupling, X_i are $N\times N$ traceless Hermitian matrices, and $A(t)$ is the gauge field. We also have periodic boundary conditions for both the scalar and gauge fields, i.e., $X^i(t+\beta) = X^i(t)$ and $A(t+\beta) = A(t)$. For this model,$d = 3$ since we have three scalar fields for the $D = 4$ model

$$X^i(t) \;\rightarrow V(t)X^i(t)V^\dagger(t), \tag{4.5}$$

$$A(t) \;\rightarrow\; V(t)(A(t) + \imath\partial_t)V^\dagger(t), \tag{4.6}$$

where $V(t) \in SU(N)$. The partition function is given by

$$Z = \int \mathscr{D}A\mathscr{D}X \exp\{-S_E\}. \tag{4.7}$$

The whole procedure of putting the bosonic BFSS model (which has an identical action as our model) on a lattice is described in [Filev 15] and [Tanwar 20]. The same arguments are used here. First, we note that here we only have temporal direction and so we will have a one-dimensional lattice and that lattice is given by $\Lambda = \{n|n = 0, 1, \ldots, T-1; t = na\}$ and imposing periodic boundary conditions implies that we need to identify 0 with T and $\beta = aT$. If we discretise the ordinary derivative we get

$$\partial_t X^i_n \rightarrow \frac{X^i_{n+1} - X^i_n}{a}. \tag{4.8}$$

To discretise the covariant derivative we need to introduce the link fields given by

$$U_{n,n+1} = \mathscr{P}\left\{\exp\left[\imath\int_{na}^{(n+1)a} dt A(t)\right]\right\} = \exp(\imath a A(t)), \tag{4.9}$$

where $\mathscr{P}$ indicates the path-ordered product. To make the above expression gauge covariant we have to transport back the field at t_{n+1} to t_n. For the discrete version of the covariant derivative, we obtain

$$\mathscr{D}_t X^i_n \rightarrow \frac{U_{n,n+1}X^i_{n+1}U^\dagger_{n,n+1} - X^i_n}{a}. \tag{4.10}$$

Substituting this back in the discretised action we get

$$S_E = \frac{Na}{2\lambda}\sum_{n=0}^{T-1}\mathrm{Tr}\left\{\left(\frac{U_{n,n+1}X^i_{n+1}U^\dagger_{n,n+1} - X^i_n}{a}\right)^2 - \frac{1}{2}[X^i_n, X^j_n]^2\right\}. \tag{4.11}$$

Simplifying the above equation we get

$$S_E = \frac{N}{\lambda}\sum_{n=0}^{T-1}\mathrm{Tr}\left\{-\frac{U_{n,n+1}X^i_{n+1}U^\dagger_{n,n+1}X^i_n}{a} + \frac{(X^i_n)^2}{a} - \frac{a}{4}[X^i_n, X^j_n]^2\right\}. \tag{4.12}$$

Now we use the $SU(N)$ symmetry to simplify the action further. If we change the fields by the following set of transformations

$$\begin{aligned}
&X^i_0 \to X^i_0,\\
&X^i_1 \to U^\dagger_{0,1}X^i_1U_{0,1},\\
&\vdots\\
&X^i_{T-1} \to (U_{0,1}U_{1,2}\ldots U_{T-2,T-1})^\dagger X^i_T(U_{0,1}U_{1.2}\ldots U_{T-2,T-1}),
\end{aligned} \tag{4.13}$$

and also if we define

$$W = (U_{0,1}U_{1,2}\ldots U_{T-2,T-1}, U_{T-1,0}), \tag{4.14}$$

we can express the action in the following way

$$S_E = -\frac{N}{a\lambda}\mathrm{Tr}\left\{WX^i_0W^\dagger X^i_{T-1} + \sum_{n=0}^{T-2}X^i_{n+1}X^i_n\right\} + \frac{N}{\lambda}\sum_{n=0}^{T-1}\mathrm{Tr}\left\{\frac{(X^i_n)^2}{a} - \frac{a}{4}[X^i_n, X^j_n]^2\right\}. \tag{4.15}$$

Upon using the result that every (finite-dimensional) Hermitian matrix can be diagonalised, we can diagonalise the W matrix by using the gauge symmetry of the field X^i_n. Let W be diagonalised by V matrix then

$$W = VDV^\dagger, \tag{4.16}$$

where $D = \mathrm{diag}(e^{\imath\theta_1}, e^{\imath\theta_2}, \ldots, e^{\imath\theta_N})$. If we transform $X^i_n \to VX^i_nV^\dagger$ then the action becomes

$$S_E = -\frac{N}{a\lambda}\mathrm{Tr}\left\{DX^i_0D^\dagger X^i_{T-1} + \sum_{n=0}^{T-2}X^i_{n+1}X^i_n\right\} + \frac{N}{\lambda}\sum_{n=0}^{T-1}\mathrm{Tr}\left\{\frac{(X^i_n)^2}{a} - \frac{a}{4}[X^i_n, X^j_n]^2\right\} \tag{4.17}$$

We will use the action for the simulation purpose. The reason we did this was to simplify the integration measure of the system. We will see that the link fields would get considerably simplified due to this choice of link fields we have taken. The integration measure for the link fields is given by

$$\prod_{n=0}^{T-1}\mathscr{D}U_{n,n+1} = \prod_{n=1}^{T-1}\mathscr{D}U_{n,n+1}\mathscr{D}U_{0,1} = \prod_{n=1}^{T-1}\mathscr{D}U_{n,n+1}\mathscr{D}W. \tag{4.18}$$

The above relation is obtained because a defining property of Haar Measure as explained in [Gattringer 07] is that being a product measure, it is cyclic and using the relation $U_{0,1} = W(U_{1,2}\dots U_{T-2,T-1})^\dagger$ we can get the above relation.

If we look carefully, the action only depends on the diagonal matrix D, which is the diagonalised form of W. So we can integrate the extra variables out and the integration measure simplifies to

$$\begin{aligned}
Z &= \int \prod_{n=0}^{T-1} \mathscr{D}U_{n,n+1}\mathscr{D}X e^{-S[X,D]} \\
&= \int \prod_{n=1}^{T-1} \mathscr{D}U_{n,n+1}\mathscr{D}W\mathscr{D}X e^{-S[X,D]} \\
&= (Vol(SU(N)))^{T-1}\int \mathscr{D}W\mathscr{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^{N} d\theta_i \prod_{j>k}^{N} \left| e^{\imath\theta_j} - e^{\imath\theta_k}\right|^2 \mathscr{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^{N} d\theta_i \prod_{j>k}^{N} \sin^2\left(\frac{\theta_j-\theta_k}{2}\right) \mathscr{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^{N} d\theta_i \mathscr{D}X e^{-S[X,D(\theta)]-S_{FP}[\theta]}.
\end{aligned} \tag{4.19}$$

The proportionality constant does not bother us because it cancels out from the numerator and the denominator when we calculate the expectation values of the observables. The Fadeev-Popov term in this equation is introduced due to the change of basis when we diagonalised W.

$$S_{FP}[\theta] = -2\sum_{j<k}^{N} \ln\left|\sin\left(\frac{\theta_j-\theta_k}{2}\right)\right| = -\sum_{j\neq k}^{N} \ln\left|\sin\left(\frac{\theta_j-\theta_k}{2}\right)\right|. \tag{4.20}$$

Since $D \in SU(N)$, all the θ variables will not be independent and is constrained by the equation

$$\sum_{i=1}^{N} \theta_i = 0, \tag{4.21}$$

as $\det(D) = 1$.

4.1.1 HMC for $D = 4$ Model

Now we will introduce the conjugate momenta for every field variables. Momentum conjugate to X_n^i is P_n^i, where P_n^i is an $N \times N$ traceless Hermitian matrix and the momentum conjugate to θ_k is P_k. The Hamiltonian for the molecular dynamics part of the HMC is

$$H = \sum_{n=0}^{T-1} \frac{1}{2}\mathrm{Tr}\left\{{P_n^i}^2\right\} + \sum_{j=1}^{N}\frac{1}{2}P_j^2 + S[X,D(\theta)] + S_{FP}[\theta]. \tag{4.22}$$

The Hamilton's equations are given by

$$\begin{aligned} \dot{X}^i_{n,rs} = \frac{\partial H}{\partial P^i_{n,sr}} = P^i_{n,rs} \quad & \dot{P}^i_{n,rs} = -\frac{\partial H}{\partial X^i_{n,sr}}, \\ \dot{\theta}_j = \frac{\partial H}{\partial P_j} = P_j \qquad & \dot{P}_j = -\frac{\partial H}{\partial \theta_j}, \end{aligned} \tag{4.23}$$

where the dot represents time derivative with respect to the fictitious time τ. These forces can be calculated using the Hamiltonian, and are given by

$$\begin{aligned} -\frac{\partial H}{\partial X^i_{0,sr}} &= \frac{N}{a\lambda}(X^i_1 - 2X^i_0 + D^\dagger X^i_{T-1} D)_{rs} + \frac{Na}{\lambda}[X^j_0,[X^i_0,X^j_0]]_{rs} \\ -\frac{\partial H}{\partial X^i_{n,sr}} &= \frac{N}{a\lambda}(X^i_{n+1} - 2X^i_n + X^i_{n-1})_{rs} + \frac{Na}{\lambda}[X^j_n,[X^i_n,X^j_n]]_{rs} \text{for } n = 1,2,\ldots,T-2 \\ -\frac{\partial H}{\partial X^i_{T-1,sr}} &= \frac{N}{a\lambda}(D^\dagger X^i_0 D - 2X^i_{T-1} + X^i_{T-2})_{rs} + \frac{Na}{\lambda}[X^j_{T-1},[X^i_{T-1},X^j_{T-1}]]_{rs} \\ -\frac{\partial H}{\partial \theta^j} &= \frac{2N}{a\lambda}\sum_{k=1}^{N} \mathrm{Re}\left(\imath X^i_{T-1,kj} X^i_{0,jk} e^{\imath(\theta_j-\theta_k)}\right)_{rs} + \sum_{k=1,k\neq j}^{N} \cot\left(\frac{\theta_j-\theta_k}{2}\right). \end{aligned} \tag{4.24}$$

4.2 Observables

The observables we are interested in calculating are the Wilson loop, Polyakov loop, the internal energy and the extent of space for the $D = 4$ model. Here we give the definition for each of them, in continuous as well as in discretised form

1. Polyakov loop $|P|$:

$$\langle |P| \rangle = \frac{1}{N}\langle |\mathrm{Tr}(P \exp^{i\int dt A(t)})| \rangle = \left\langle \left| \frac{\mathrm{Tr}(D)}{N} \right| \right\rangle \text{ (On Lattice)}$$

2. Extent of space R^2: The extent of space can be regarded as the gauge invariant version of the region of the configuration space covered by the bosonic fields.

$$\langle R^2 \rangle = \frac{\lambda}{N\beta}\left\langle \int_0^\beta dt \mathrm{Tr}(X^{i^2}) \right\rangle = \frac{\lambda}{NT}\left\langle \sum_{n=0}^{T-1} dt \mathrm{Tr}(X^{i^2}_n) \right\rangle \text{ (On Lattice)}$$

3. Internal energy E: Internal energy is the total energy of the system and is used for determining the order of the phase transition [Hanada 07].

$$\langle \frac{E}{N^2} \rangle = -\frac{3\lambda}{4N\beta}\left\langle \int_0^\beta dt \mathrm{Tr}([X^i,X^j]^2) \right\rangle$$

$$= -\frac{3\lambda}{4NT}\left\langle \sum_{n=0}^{T-1} dt \mathrm{Tr}([X^i_n,X^j_n]^2) \right\rangle \text{ (On Lattice)}$$

4. Wilson loop (See 3)

$$\langle |W| \rangle = \left\langle \left| \frac{1}{N} \mathrm{Tr} \mathscr{P} \exp \left(\int_0^\beta dt (\imath A(t) + X_3(t)) \right) \right| \right\rangle$$

$$= \left\langle \left| \frac{1}{N} \mathrm{Tr} \mathscr{P} \exp \left(\sum_0^N a (\imath A(t) + X_3(t)) \right) \right| \right\rangle \text{(On Lattice)}$$

4.3 Simulation Results for $D = 4$ Model

4.3.1 Simulation Details

For our simulation in $T > 0.2$, as in Ref. [Tanwar 20], we changed the lattice spacing a to change the temperature while keeping the lattice size T fixed. Also, we opted for a large lattice size deliberately to keep the lattice spacing small, as large values of a would greatly increase the time taken for the simulation. However for low temperatures $T < 0.2$, we had to increase the lattice size as the system did not stabilise with the given lattice size. The lattice size we used is shown in Tab. 4.1.

Temp	Spacing	No. of Sites
0.12	0.3333	25
0.16	0.3125	20
0.20	0.5000	10
0.30	0.3333	10
0.40	0.2500	10
0.50	0.2000	10
0.60	0.1667	10
0.65	0.1538	10
0.85	0.1176	10
1.00	0.1000	10
1.15	0.0869	10
1.20	0.0833	10
1.40	0.0714	10
1.80	0.0556	10
2.10	0.0476	10

Table 4.1: Lattice spacing and the number of lattice sites for each temperature.

We fixed the following values for every simulation: $d = 3$, $N = 4$, $\lambda = 1.0$, and $T = 15$. The total number of sweeps made was 100000 and the thermalization was done for 20000 sweeps. Since we are dealing with a large number of sweeps and the computation took quite a while, the code was written in **bash** script to ensure that the field configuration was saved after every 200 sweeps. To make the computation faster, we ran the simulations for different temperatures in different cores simultaneously. The temperature varies from 0.12 to 2.1.

In the simulations, each variable and parameter has been measured in unit of 't Hooft coupling λ_0. Since the Yang-Mills coupling g^2 has a dimension of $[L^{-3}]$ in one dimension, so X_i^n and A^n can be measured in the unit of $\lambda_0^{1/3}$ and a can be measured in units of $\lambda_0^{-1/3}$. Thus all terms in the action Eq. (4.4) are dimensionless.

4.3.2 Results

The Wilson loop given in Fig. 4.1 does not follow any particular trend. It is almost constant and non-zero throughout the temperature range. We did not find any similar plots during our literature survey. However, keeping in mind that the Polyakov loop is a special case of the Wilson loop, and it shows the transition as in Fig. 4.4, one could expect a similar kind of transition in the Wilson loop also. However, the non-zero values for the Wilson loop operator means that the coupling between the gauge and the field variable is non-zero.

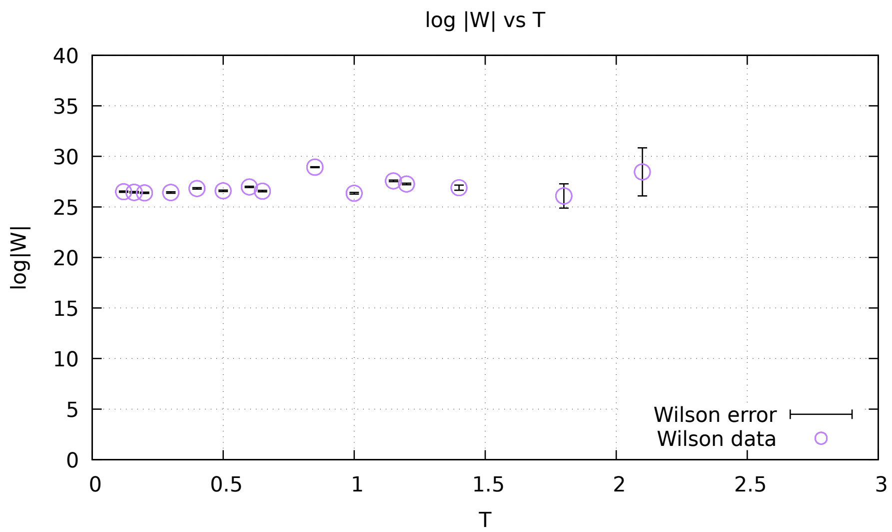

Figure 4.1: Wilson loop against T in the $D = 4$ model.

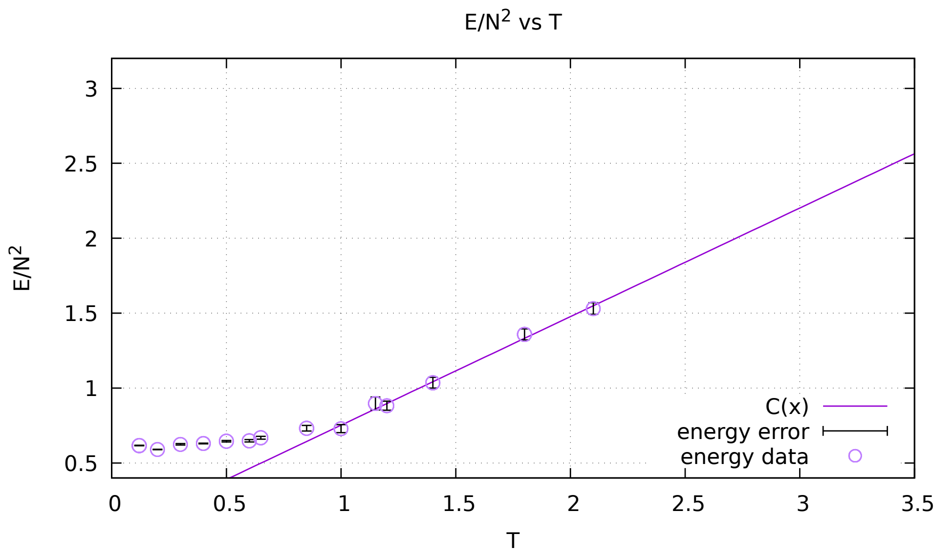

Figure 4.2: Internal energy against T in the $D = 4$ model.

Figure 4.2 shows the plot of the scaled internal energy against temperature. We see that there is a transition at around $T = 0.8$, which is expected since there is a phase transition at that temperature. For large values of N, we would have noticed a sharp kink. Thus we could have confirmed that it was indeed a first-order phase transition.

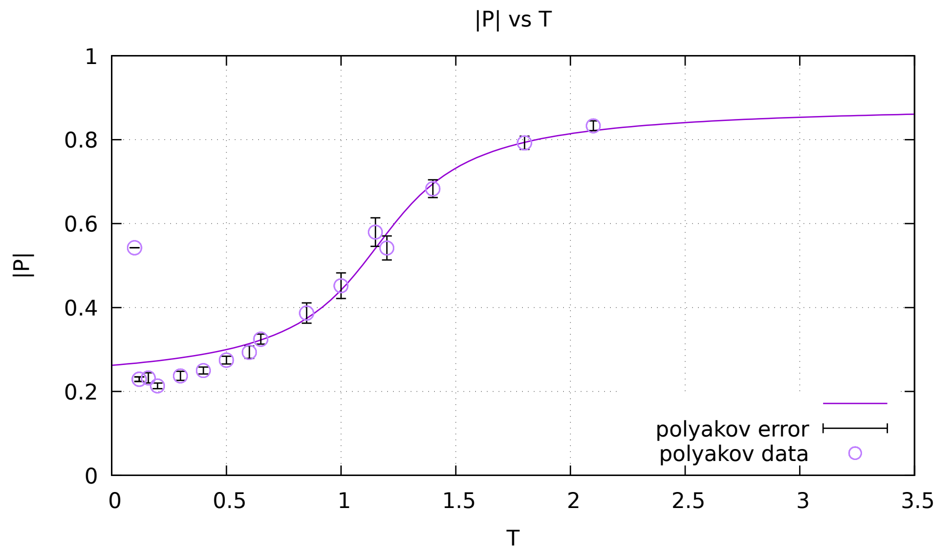

Figure 4.3: Polyakov loop against T in the $D = 4$ model. Here the purple line shows the fitted curve.

We can see a clear phase transition by looking at the Polyakov loop operator in Fig. 4.3. Our results are in excellent agreement with Ref. [Tanwar 20]. We fitted the Polyakov loop using the equation $|P(T)| = A\arctan(B(T-C)) + D$ and the fitting parameters are listed below.

Fitting Parameter	Value
A	0.29(78)
B	1.9(92)
C	1.1(13)
D	0.54(53)

Table 4.2: Table showing the fitting parameters for Polyakov loop.

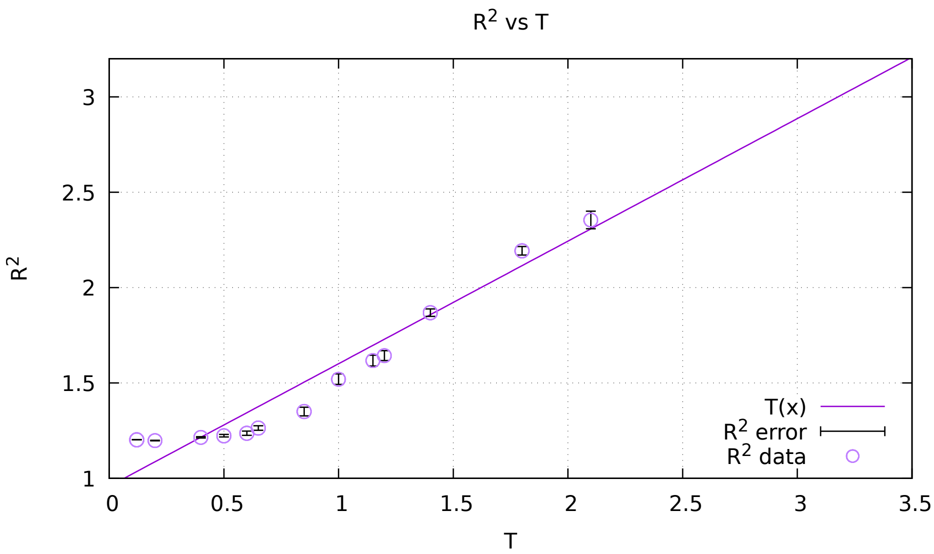

Figure 4.4: Extent of space against T in the $D = 4$ model.

The extent of space in Fig. 4.4 shows a transition at around $T = 0.8$ after which it can be approximated as a straight line which is given by the purple line in the figure. This is a test for ensuring that the HMC algorithm does not suffer from flat directions.

4.4 BFSS Model

The action of the BFSS model is

$$\begin{aligned} S &= \frac{1}{g^2}\int_0^\beta dt \ \ \mathrm{Tr}\Big[(D_tX_i)^2 - \frac{1}{2}[X_i,X_j]^2 - \imath\Psi_\alpha^T C_{10}\gamma^0_{\alpha\beta}D_t\Psi_\beta \\ &\qquad + (C_{10}\gamma^i_{\alpha\beta}[X_i,\Psi_\beta]\Big]. \end{aligned} \tag{4.25}$$

4.4.1 Bosonic BFSS

The procedure of putting the bosonic BFSS on the lattice follows a similar procedure as that of the $D = 4$ model, except that in this case, the number of scalars involved is different.

Thus the Euclidean action for the bosonic BFSS is

$$S_E = \frac{N}{\lambda}\int_0^\beta dt \;\; \mathrm{Tr}\left[\frac{1}{2}(D_t X_i)^2 - \frac{1}{4}[X_i, X_j]^2\right], \tag{4.26}$$

while the action that we use for simulations is

$$S_E = -\frac{N}{a\lambda}\mathrm{Tr}\left\{DX_0^i D^\dagger X_{T-1}^i + \sum_{n=0}^{T-2} X_{n+1}^i X_n^i\right\} + \frac{N}{\lambda}\sum_{n=0}^{T-1}\mathrm{Tr}\left\{\frac{(X_n^i)^2}{a} - \frac{a}{4}[X_n^i, X_n^j]^2\right\}. \tag{4.27}$$

Observables

The observables we are interested in calculating are the Wilson loop, Polyakov loop, the internal energy and the extent of space for the bosonic BFSS model, similar to the $D = 4$ model, with one notable exception i.e., the Wilson loop, which is coupled to the last scalar, which is the 9th in this case. However, it is expected that due to the $SU(N)$ symmetry, that any scalar we choose would not affect the final result. We checked it for the 3rd and the 9th scalars for two temperatures, and the values did not differ.

1. Polyakov Loop $|P|$:

$$\langle |P| \rangle = \frac{1}{N}\langle |\mathrm{Tr}(P\exp^{i\int dtA(t)})| \rangle = \left\langle \left| \frac{\mathrm{Tr}(D)}{N} \right| \right\rangle \text{(On Lattice)}$$

2. Extent of space R^2:

$$\langle R^2 \rangle = \frac{\lambda}{N\beta}\left\langle \int_0^\beta dt \mathrm{Tr}(X^{i^2}) \right\rangle = \frac{\lambda}{NT}\left\langle \sum_{n=0}^{T-1} dt \mathrm{Tr}(X_n^{i^2}) \right\rangle \text{(On Lattice)}$$

3. Internal energy E

$$\left\langle \frac{E}{N^2} \right\rangle = -\frac{3\lambda}{4N\beta}\left\langle \int_0^\beta dt \mathrm{Tr}([X^i, X^j]^2) \right\rangle$$

$$= -\frac{3\lambda}{4NT}\left\langle \sum_{n=0}^{T-1} dt \mathrm{Tr}([X_n^i, X_n^j]^2) \right\rangle \text{(On Lattice)}$$

4. Wilson loop (See 3)

$$\langle |W| \rangle = \left\langle \left| \frac{1}{N}\mathrm{Tr}\mathscr{P}\exp\left(\int_0^\beta dt(\imath A(t) + X_9(t))\right) \right| \right\rangle$$

$$= \left\langle \left| \frac{1}{N}\mathrm{Tr}\mathscr{P}\exp\left(\sum_0^N a(\imath A(t) + X_9(t))\right) \right| \right\rangle \text{(On Lattice)}$$

HMC for Bosonic BFSS

The force equations are exactly same as that of Eq. (4.24) since we have only changed the number of scalars.

Simulation Details

For our simulation with $T > 0.2$, as in Ref. [Tanwar 20], we changed the lattice spacing a to change the temperature while keeping the lattice size T fixed. Also, we opted for a large lattice size deliberately to keep the lattice spacing small, as large values of a would greatly increase the time taken for the simulation. The lattice size we used is shown in Tab. 4.1.

We fixed the following values for every simulation: $d = 9$, $N = 4$, $\lambda = 1.0$, and $T = 15$. The total no of sweeps made was 600000 and the thermalisation was done for 100000 sweeps. Since we are dealing with a large number of sweeps and the computation took quite a while, the code was written in **bash** script to ensure that the field configuration was saved after every 200 sweeps. To make the computation faster, we ran the simulations for different temperatures in different cores simultaneously. The temperature varies from 0.20 to 2.4.

In the simulations, each variable and parameter has been measured in unit of the 't Hooft coupling λ_0. Since the Yang-Mills coupling g^2 has a dimension of $[L^{-3}]$ in one dimension, so X_i^n and A^n can be measured in unit of $\lambda_0^{1/3}$ and a can be measured in unit of $\lambda_0^{-1/3}$. Thus all terms in the action Eq. 4.26 are dimensionless.

Simulation Results

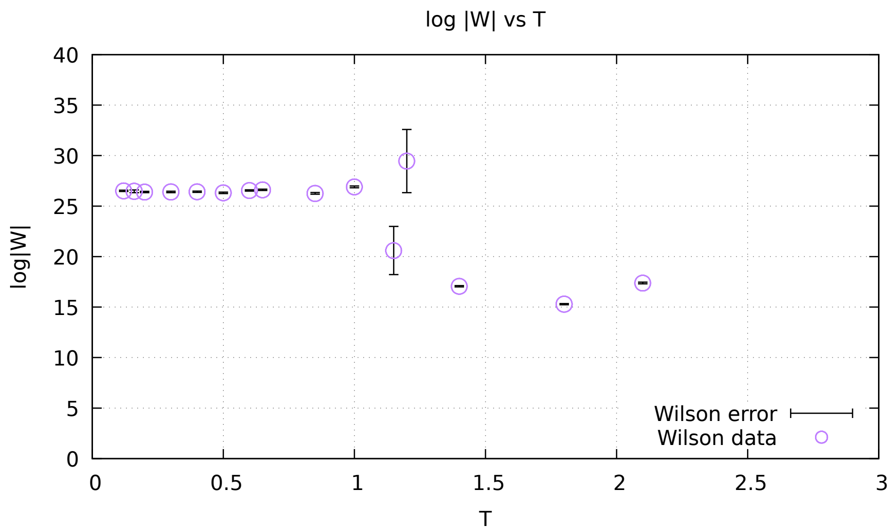

Figure 4.5: Wilson loop against T in the bosonic BFSS model

Here the Wilson loop in Fig. 4.5 shows a clear transition above $T = 1.0$. This indicates a very steep phase transition. We can also understand that there is a phase transition because, in these regions, the error in the expected value is much larger than in other temperatures. Again, we did not find any similar study during our literature survey, but one must note that we did not obtain any transition during our simulations on the $D = 4$ model. Figure 4.5 does not bear any resemblance to the results for the full model as given in Ref. [Nishimura 07].

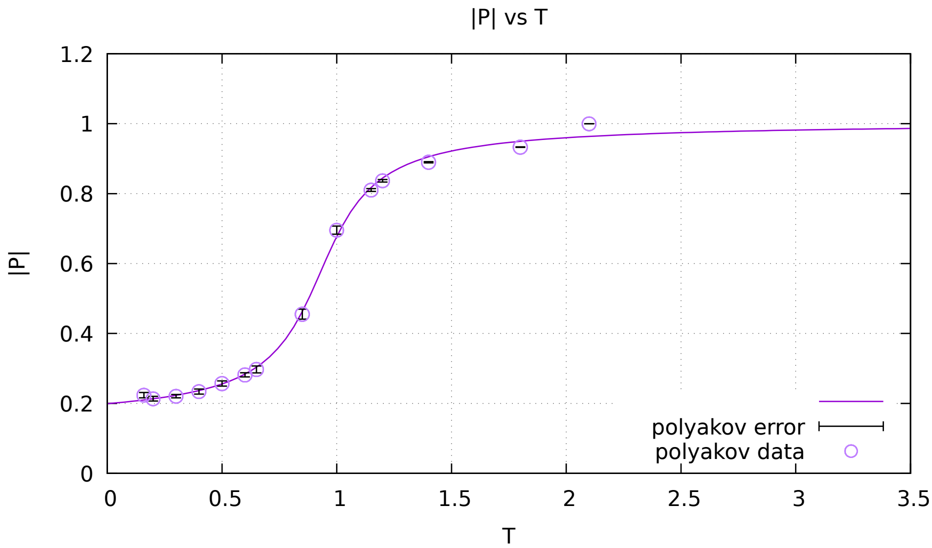

Figure 4.6: Polyakov loop against T in the bosonic BFSS model. Here also, the purple line shows the fitted curve.

The Polyakov loop transition in Fig. 4.6 is much steeper as compared to the $D = 4$ model (Fig. 4.3). It is expected since the Polyakov loop is the order parameter, so more the number of scalars, the more would be the disorder in the system during the confined-deconfined phase transition. Here we fitted the Polyakov loop against temperature plot using the equation $|P(T)| = A \arctan(B(T - C)) + D$ and the fit values obtained are listed in the below table.

Fitting Parameter	Value
A	0.27(73)
B	5.5(51)
C	0.92(10)
D	0.57(60)

Table 4.3: Table showing the fit parameters for the Polyakov loop.

Here, C indicates the critical temperature at which the confined- deconfined transition takes place. Here $C = 0.927 \pm 0.082$ was obtained after fitting the curve.

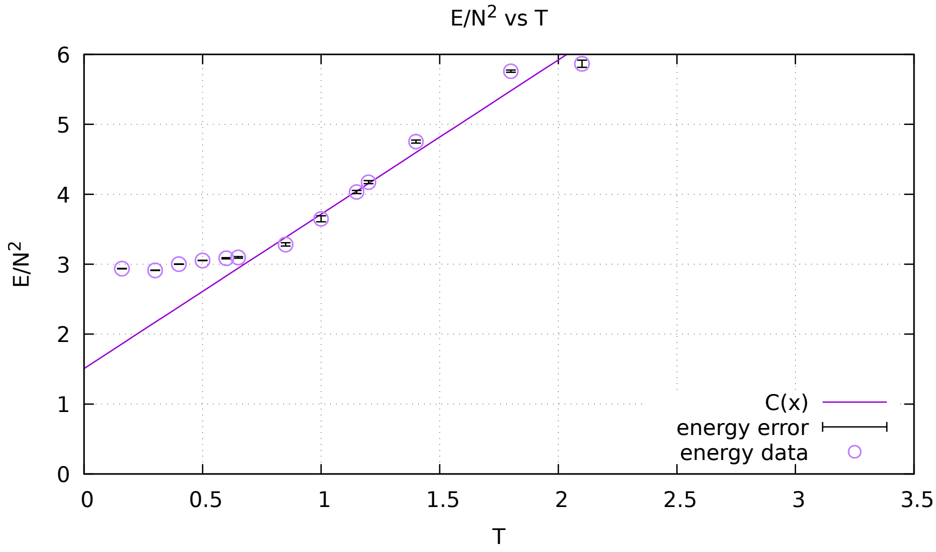

Figure 4.7: Internal energy against T in the bosonic BFSS model.

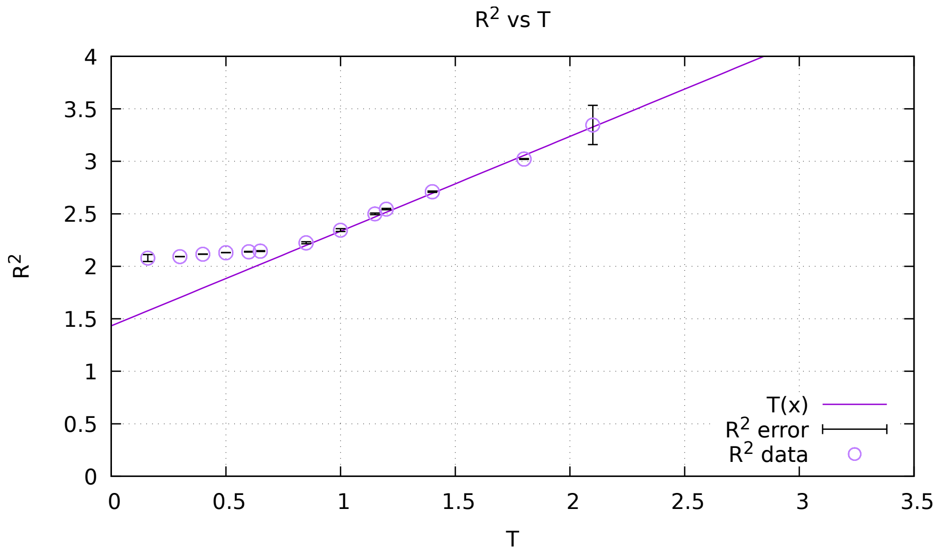

Figure 4.8: Extent of space against T in the bosonic BFSS model.

The scaled internal energy given in Fig. 4.7 and the extent of space in Fig. 4.8 show that these observables are similar to those obtained in Ref. [Tanwar 20]. It is interesting to note that both the extent of space and internal energy has higher values corresponding to the same temperature compared with the $D = 4$ model. That is due to more number of scalars involved in the bosonic BFSS.

4.4.2 The BFSS Model

Due to lack of time, we could not simulate the full supersymmetric BFSS model on the lattice and hence could not compute the Schwarzschild radius. However, here we lay down the steps involved. The discretisation of the full BFSS matrix model is worked out in detail in Appendix A.

We can write the fermionic part of the action as

$$S_f = \psi^T \mathscr{M} \psi. \tag{4.28}$$

Integrating out the fermions from the action, the partition function can be written as

$$Z \propto \int \mathscr{D}X \mathscr{D}\psi e^{-S[X,\psi]} \propto \int \mathscr{D}X \mathrm{Pf}(\mathscr{M}) e^{-S_{bos}[X]}. \tag{4.29}$$

Since $\mathscr{M}$ is the sum of the kinetic term and the anti-Hermitian potential term, we have $\mathscr{M}^\dagger(x) = -\mathscr{M}(-x)$. Also because the bosonic action is symmetric in X the Pfaffian in the partition function can be replaced by $|\mathrm{Pf}(\mathscr{M})|\cos\big(\theta_{Pf}\big)$. Now as long as $-\frac{\pi}{2} < \Theta_{Pf} < \frac{\pi}{2}$ the cosine is positive and the effective action defines a true probability distribution given by

$$S_{eff} = S_{bos}[X] - \ln|\mathrm{Pf}(\mathscr{M})| - \ln\cos\big(\Theta_{Pf}\big). \tag{4.30}$$

Since the cosine remains positive [Filev 15], the drop in the $\cos\theta$ curve at very low temperatures might be a lattice effect and is not present in the continuum limit.

Thus we can write the Pfaffian as

$$|\mathrm{Pf}\mathscr{M}| = \det(\mathscr{D}^{\frac{1}{4}}) \text{ where } \mathscr{D} = \mathscr{M}^\dagger\mathscr{M}. \tag{4.31}$$

It is evident from the above expression that exact computation of the Pfaffian is a computationally inefficient process. So we resort to the rational HMC algorithm to help us in this situation. The idea of the RHMC algorithm is to use the approximate the rational exponent of of the matrix $\mathscr{M}^\dagger\mathscr{M}$ with a partial sum:

$$(\mathscr{M}^\dagger\mathscr{M})^\delta = \alpha_0 + \sum_{i=1}^{Q} \alpha_i(\mathscr{M}^\dagger\mathscr{M} + \beta_i)^{-1}, \tag{4.32}$$

which has sufficiently small error within a certain range required by the system to be simulated and hence we can use a partial sum. The real positive parameters α_i and β_i are obtained by a code based on the Remez algorithm [Clarke 05] (the parameters α_0; α_i; β_i and Q depend on the rational exponent δ, the spectral range of the matrix $\mathscr{M}^\dagger\mathscr{M}$ and the desired accuracy).

We will need two rational exponents for computing the forces. In order to update the pseudo fermions we use that the field $\eta \equiv (\mathscr{M}^\dagger\mathscr{M})^{-\frac{1}{8}}F$ has a Gaussian distribution and solve for $\varepsilon = (\mathscr{M}^\dagger\mathscr{M})^{\frac{1}{8}}\eta$ using a multi-shift solver [Jegerlehner 96]. Therefore, $\delta = \frac{1}{8}$ is one of the rational exponents that we need. To calculate the fermionic forces and the contribution to the Hamiltonian we need to invert $(\mathscr{M}^\dagger\mathscr{M})^{-\frac{1}{4}}$ and the second exponent is $\delta = -\frac{1}{4}$.

As the model does not suffer from a severe sign problem, we can ignore the phase of the Pfaffian and use Eq. (4.31) to write

$$S_{PF} \equiv F^\dagger(\mathscr{M}^\dagger\mathscr{M})^{-\frac{1}{4}}F, \tag{4.33}$$

where

$$Z \propto \int \mathscr{D}X\mathscr{D}F^\dagger\mathscr{D}Fe^{-S_{bos}[X]-S_{PF}}. \tag{4.34}$$

Then the Pfaffian is replaced by

$$|\mathrm{Pf}\mathscr{M}| = \int dFdF^* e^{-S_{PF}} \text{ given } S_{PF} = \alpha_0 F^*F + \sum_{i=1}^{Q}\alpha_i F^*(\mathscr{D}+\beta_i)^{-1}F, \tag{4.35}$$

using the auxiliary complex variables F, which is called the pseudo-fermions. Here F is a $16(N^2-1)\Lambda$ dimensional vector containing the pseudo-fermionic fields where N is the number of colours used. To this end we need the so-called fermionic forces.

We have two type of forces - gradients with respect to $X_{n,ij}$ and gradients with respect to the gauge variables θ_i. Using the partial expansion Eq. (4.32), one can easily derive expression for the derivatives of the fermionic action:

$$\begin{aligned} \frac{\partial S_{PF}}{\partial u} &= -\textstyle\sum_{i=1}^{Q}\alpha_i F^\dagger(\mathscr{M}^\dagger\mathscr{M}+\beta_i)^{-1}\frac{\partial(\mathscr{M}^\dagger\mathscr{M})}{\partial u}(\mathscr{M}^\dagger\mathscr{M}+\beta_i)^{-1}F \\ &= -\textstyle\sum_{i=1}^{Q}\alpha_i {h_i}^\dagger\frac{\partial(\mathscr{M}^\dagger\mathscr{M})}{\partial u}h_i \end{aligned} \tag{4.36}$$

where h_i satisfy $(\mathscr{M}^\dagger\mathscr{M}+\beta_i)h_i = F_i$ and are obtained from the multi-solver.

For calculating the forces, we use the following formulae

$$\frac{d\phi'_n(\tau)}{d\tau} = \alpha_n\frac{\partial S_{HMC}}{\partial\Pi'_n} = \alpha_n\Pi'_{-n}, \tag{4.37}$$

$$\frac{d\Pi'_n(\tau)}{d\tau} = -\alpha_n\frac{\partial S_{HMC}}{\partial\phi'_n}. \tag{4.38}$$

The real coefficients α_n should be optimised based on the idea of the Fourier acceleration [Catterall 08]. This evolution, if treated exactly, conserves the action S_{HMC}. In practice, we discretise the τ-evolution in a way (the leapfrog discretisation) that the reversibility is not affected. However, since during evolution τ is not arbitrarily small, the action changes. Let us say that the action S_{HMC} changes by a small amount ΔS_{HMC}. In order to satisfy the detailed balance, we accept the new configuration with the probability minimum of 1 and $e^{-\Delta S_{HMC}}$. Before we start the next τ-evolution, we refresh the Π'_n variables by drawing

Gaussian random numbers, which follows from the action. This procedure is necessary for avoiding the ergodicity problem. (The step size $\Delta\tau$ should be optimised for fixed τ_f by maximising the acceptance rate times $\Delta\tau$. Then τ_f should be optimised by minimising the autocorrelation time in units of step in the τ-evolution).

The expected plot for Wilson loop against temperature is shown in Fig. 4.9.

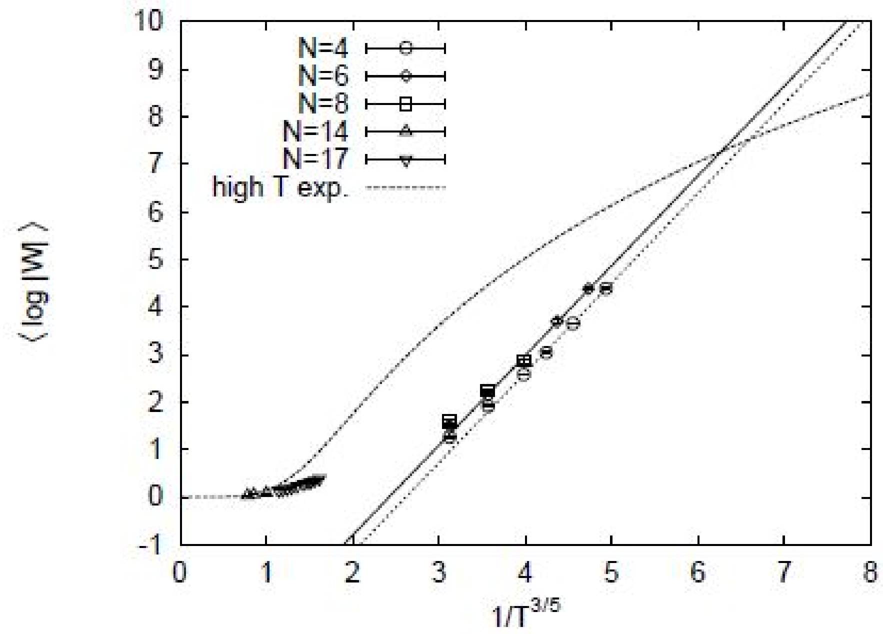

Figure 4.9: The expected value of the Wilson loop as given in Ref. [Nishimura 07].

We expect to obtain these results on using the above procedure on the full BFSS Model. Once we obtain this result, then the Schwarzschild radius can also be computed.

Chapter 5

Summary

In this thesis, we have tried to briefly explain the rationale and the steps involved in the Markov Chain Monte Carlo (MCMC) algorithm, which is the computational technique we have used in our subsequent work. We then briefly explained the principal calculation involved in showing the equivalence between the Wilson loop operator with the Schwarzschild radius. Then, to test the effectiveness of the Monte Carlo routine, we implemented it for the $D = 4$ model and after that, we calculated the Wilson loop for the bosonic BFSS model and gave the steps involved in implementing the RHMC for the full BFSS model.

Future work on this project would be testing the validity of the results given in Ref. [Nishimura 07]. If the results agree, then this would give us a powerful tool for analysing strongly coupled gauge theories, which cannot be studied perturbatively and thus this method would be useful in verifying theoretical predictions on the gauge theory side.

Appendix A

Discretisation of Fermionic Systems

The action for the full BFSS matrix model is obtained after dimensionally reducing the $9+1$-dimensional supersymmetric Yang-Mills theory down to $0+1$-dimensions. The action is given by

$$S = \frac{1}{g^2}\int_0^\beta dt \ \mathrm{Tr}\Big[(D_t X_i)^2 - \frac{1}{2}[X_i, X_j]^2 - \imath \Psi_\alpha^T C_{10} \gamma^0_{\alpha\beta} D_t \Psi_\beta + C_{10}\gamma^i_{\alpha\beta}[X_i, \Psi_\beta]\Big]. \qquad \text{(A.1)}$$

A naive discretisation of the action could result in fermion doubling. This arises mainly in lattice formulation due to multiple singularities given the periodic boundary conditions [Gattringer 07]. This can be avoided if the charge conjugation matrix is taken to be $C_9 = I_8 \otimes \sigma_1$. Constructing a basis for which C_9 is in this form is straight-forward. For example one can tensor up the Majorana basis in 7 Euclidean dimensions $\gamma_E'^a$

$$\begin{aligned} \gamma^a &= -\gamma_E'^a \otimes \sigma_3, \ \mathrm{a} = \{1, \ldots, 7\} \\ \gamma^8 &= 1_8 \otimes \sigma_2 \\ \gamma^9 &= 1_8 \otimes \sigma_1 \end{aligned} \qquad \text{(A.2)}$$

and verify that indeed C_9 is of the desired form (it also satisfies $C_9 = \gamma^9$).

The Gamma Matrices in full are given as

$$\gamma_1 = \imath\sigma_2 \otimes \sigma_2 \otimes \sigma_2 \otimes \sigma_2; \gamma_2 = \imath\sigma_2 \otimes \sigma_2 \otimes \mathbf{1} \otimes \sigma_1; \gamma_3 = \imath\sigma_2 \otimes \sigma_2 \otimes \mathbf{1} \otimes \sigma_3;$$

$$\gamma_4 = \imath\sigma_2 \otimes \sigma_1 \otimes \sigma_2 \otimes \mathbf{1}; \gamma_5 = \imath\sigma_2 \otimes \sigma_3 \otimes \sigma_2 \otimes \mathbf{1}; \gamma_6 = \imath\sigma_2 \otimes \mathbf{1} \otimes \sigma_1 \otimes \sigma_2$$

$$\gamma_7 = \imath\sigma_2 \otimes \mathbf{1} \otimes \sigma_3 \otimes \sigma_2; \gamma_8 = \imath\sigma_1 \otimes \mathbf{1} \otimes \mathbf{1} \otimes \mathbf{1}; \gamma_9 = \imath\sigma_3 \otimes \mathbf{1} \otimes \mathbf{1} \otimes \mathbf{1}$$

$$\gamma_{10} = \imath\mathbf{1} \otimes \mathbf{1} \otimes \mathbf{1} \otimes \mathbf{1}$$

In this representation, the charge conjugation matrix becomes the identity matrix [Ambjørn 00].

A.1 Lattice Formulation

The procedure detailed below is given in Ref. [Filev 15]. We are interested in discretising the fermionic part of the action since the bosonic part is easily obtained using simple discretistaion involving the gauge link fields (See 4):

$$S_f = \frac{1}{2g^2}\int d\tau \mathrm{Tr}\left(\psi^\alpha (C_9)_{\alpha\beta}\mathscr{D}_\tau\psi^\beta - \psi^\alpha(C_9\gamma^i)_{\alpha\beta}[X^i,\psi^\beta]\right). \tag{A.3}$$

We begin by splitting the fermions into two eight component fermions $\psi = (\psi^1, \psi^2)$ (analogous to using staggered fermions, which in one dimension completely removes the doublers) and defining the forward and backward derivatives D$\pm$:

$$\begin{aligned}(\mathrm{D}_-W)_n &= \frac{(W_n - U_{n,n-1}W_{n-1}U_{n-1,n})}{a}\\ (\mathrm{D}_+W)_n &= \frac{(U_{n,n+1}W_{n+1}U_{n+1,n} - W_n)}{a}\end{aligned} \tag{A.4}$$

One can show that the discretised kinetic term then becomes:

$$\begin{aligned}S^f_{kin} &= \frac{1}{2g^2}\int d\tau \mathrm{Tr}\left(\psi^\alpha (C_9)_{\alpha\beta}\mathscr{D}_\tau\psi^\beta\right)\\ &= \frac{a}{2g^2}\sum_{n=1}^{\Lambda-1}\mathrm{Tr}\left(\psi^T_{1,n}(\mathrm{D}_-\psi_2)_n + \psi^T_{2,n}(\mathrm{D}_+\psi_1)_n\right)\\ &= \frac{1}{g^2}\left(-\sum_{n=0}^{\Lambda-1}\psi^T_{2,n}\psi_{1,n} + \sum_{n=0}^{\Lambda-2}\psi^T_{2,n}U_{n,n+1}\psi_{1,n+1}U_{n+1,n} \pm \psi^T_{2,\Lambda-1}U_{\Lambda-1,0}\psi_{1,0}U_{0,\Lambda-1}\right),\end{aligned} \tag{A.5}$$

where the plus / minus sign in the last term corresponds to periodic / anti-periodic boundary conditions for the fermions. Using the gauge from the bosonic part when the holonomy is concentrated on a singe link we can write S^f_{kin} as

$$S^f_{kin} = \frac{1}{g^2}\left(-\sum_{n=0}^{\Lambda-1}\psi^T_{2,n}\psi_{1,n} + \sum_{n=0}^{\Lambda-2}\psi^T_{2,n}\psi_{1,n+1} \pm \psi^T_{2,\Lambda-1}\mathrm{D}\psi_{1,0}\mathrm{D}^\dagger\right). \tag{A.6}$$

Since all fields transform in the adjoint of $SU(N)$ instead of dealing with matrices we can use the corresponding real components: $X_a = \mathrm{Tr}(t_a X)$, where t_a are the standard basis of $SU(N)$ (for $N = 4$, in our case) normalised as $\mathrm{Tr}(t_a\ t_b) = \delta_{ab}$. Then S^f_{kin} can then be written as:

$$S^f_{kin} = \frac{1}{g^2}\sum_{a,b=0}^{N^2-1}\sum_{m,n=0}^{\Lambda-1}\sum_{\alpha,\beta=1}^{16}\psi^{\alpha+8}_{m,a}K^{ab}_{mn}\psi^\alpha_{n,b}, \tag{A.7}$$

$$K^{ab}_{mn} = (\delta_{m+1,n} - \delta_{m,n})\delta^{ab} \pm \delta_{m,\Lambda-1}\delta_{n,0}d^{ab}, \tag{A.8}$$

$$d^{ab} = \mathrm{Tr}(t^a D t^b D^\dagger), \tag{A.9}$$

where the plus/minus sign corresponds to periodic/ant-periodic boundary conditions. The kinetic term can also be written as:

$$S^f_{kin} = \sum_{a,b=0}^{N^2-1} \sum_{m,n=0}^{\Lambda-1} \sum_{\alpha,\beta=1}^{16} \psi^{\alpha}_{m,a} \mathscr{M}^{kin}_{mn,\alpha\beta,ab} \psi^{\beta}_{n,b}, \tag{A.10}$$

$$\mathscr{M}^{kin}_{mn,\alpha\beta,ab} = \frac{1}{2g^2} \begin{pmatrix} 0_8 & -K^{ba}_{nm} \\ K^{ab}_{mn} & 0_8 \end{pmatrix}_{\alpha\beta}. \tag{A.11}$$

Discretising the potential part of the action is straightforward. One obtains:

$$S^f_{pot} = \sum_{a,b=0}^{N^2-1} \sum_{m,n=0}^{\Lambda-1} \sum_{\alpha,\beta=1}^{16} \psi^{\alpha}_{m,a} \mathscr{M}^{pot}_{mn,\alpha\beta,ab} \psi^{\beta}_{n,b}, \tag{A.12}$$

$$\mathscr{M}^{pot}_{mn,\alpha\beta,ab} = \frac{1}{2g^2} a\imath f^{abc} (C_9 \gamma^i)_{\alpha\beta} X^{c,i}_n \delta_{n+m,0}. \tag{A.13}$$

Finally, defining:

$$\mathscr{M}_{mn,\alpha\beta,ab} = \frac{1}{2g^2} \begin{pmatrix} 0_8 & -K^{ba}_{nm} \\ K^{ab}_{mn} & 0_8 \end{pmatrix}_{\alpha\beta} + \frac{1}{2g^2} a\imath f^{abc} (C_9 \gamma^i)_{\alpha\beta} X^{c,i}_n \delta_{n+m,0}. \tag{A.14}$$

Thus the fermionic part of the action can be written as:

$$S_f = \psi^T \mathscr{M} \psi. \tag{A.15}$$

A.2 Non-Lattice Formulation

The importance of the lattice formulation lies in its manifest gauge invariance [Filev 15]. In the 1-dimensional case, however, the gauge dynamics is almost trivial. This allows us to use a non-lattice formulation. A non-lattice simulation is computationally more efficient compared to the lattice formulation, however, it can be effectively used only in the 1-dimensional case.

Essentially, one can take a static diagonal gauge and using the residual large gauge transformation, we can choose a gauge slice such that the diagonal elements of the constant gauge field lie within a minimum interval. Finally, we have to expand the fields into Fourier modes and keep only the modes up to some cutoff. This is specific to one dimension, and the momentum cutoff regularisation in higher dimensions generally breaks gauge invariance [Hanada 07].

Let us first analyze the bosonic part of BFSS whose action is given by

$$S_b = \frac{N}{2\lambda} \int_0^{\beta} dt \ \mathrm{Tr} \left((D_t X_i)^2 - \frac{1}{2} [X_i, X_j]^2 \right). \tag{A.16}$$

We take a static diagonal gauge of the form

$$A(t) = \frac{1}{\beta}(\alpha_1, \alpha_2, \dots, \alpha_N) \tag{A.17}$$

and introduce a cutoff Λ while calculating the Fourier modes of the scalar fields

$$X_i^{ab} = \sum_{n=-\Lambda}^{n=\Lambda} X_{in}'^{ab} e^{\imath n\omega t}. \tag{A.18}$$

Due to the residual gauge symmetry, we have the following conditions (where $g(t) = (\exp(\imath\omega\nu_1 t), \dots, \exp(\imath\omega\nu_N t)))$

$$X_i \to gX_ig^\dagger \implies X'_{(i,n)} = X'_{(i,n-\nu_a+\nu_b)} \tag{A.19}$$

$$A \to gAg^\dagger + \imath g\partial_t g^\dagger \implies \alpha_a = \alpha_a + 2\pi\nu_a, \tag{A.20}$$

which is fixed by imposing the following condition

$$-\pi < \alpha_a < \pi \tag{A.21}$$

This ensures that the gauge dynamics are restricted within an interval. Thus we decompose the scalar fields into the Fourier modes and introduce a cutoff Λ. This momentum cutoff regularization is valid only in the present one-dimensional case since in higher dimensions, it breaks gauge invariance.

$$X_i^{ab} = \sum_{n=-\Lambda}^{n=\Lambda} X_{in}'^{ab} e^{\imath n\omega t}, \tag{A.22}$$

where $\omega = \frac{2\pi}{\beta}$ which comes from the periodicity condition $X_i(t+\beta) = X_i(t)$. (Here the upper indices indicate the matrix element.) Since X_i are traceless, Hermitian $N \times N$ matrices, we get the condition that the Fourier modes X'_{ni} are traceless $N \times N$ matrices satisfying the following constraint

$$X_{ni}'^{\dagger} = X'_{-ni}. \tag{A.23}$$

In the final action, we need to introduce the Fadeev-Popov term since we are considering a static diagonal gauge

$$S_{FP}[\alpha] = -2\sum_{j<k}^{N} \ln\left|\sin\left(\frac{\alpha_j - \alpha_k}{2}\right)\right|. \tag{A.24}$$

While calculating the Fourier transformed action, we need only concern ourselves with the cases where the exponential has zero phase. So by integrating out the time-dependent part we are left with the following action

$$S_b = \frac{N}{2\lambda}\sum_{n=-\Lambda}^{\Lambda}\sum_{a,k=1}^{N}\left(n\omega - \frac{\alpha_a - \alpha_b}{\beta}\right)^2 \beta X_{ni}'^{ab} X_{-ni}'^{ba} - \frac{N\beta}{4\lambda}\operatorname{Tr}\left(\sum_{k=-\Lambda}^{\Lambda}[X'_{ni}, X'_{mj}][X'_{oi}, X'_{-ki}]\right)$$

$$k = n + m + o \,, p = -k. \tag{A.25}$$

For the HMC algorithm, the Hamiltonian will be

$$H = S_b + S_{FP}[\alpha] + \sum_{j=0}^{N} \frac{1}{2} p_j^2 + \sum_{j=0}^{N} \mathrm{Tr}\left(\frac{1}{2} \sum_{n=-\Lambda}^{\Lambda} \Pi_{-nj} \Pi_{nj} \right). \tag{A.26}$$

Here Π_i are the $N \times N$ Hermitian, traceless matrices which are the conjugate momentum variables to X_i, whereas the p_i are N real variables conjugate variables to α_i. The force due to the Fadeev-Popov terms α_a is given by

$$-\frac{\partial H}{\partial \alpha_a} = \sum_{a \neq b} \left[\sum_{n=-\Lambda}^{\Lambda} \frac{2N}{\beta} \left(n\omega - \frac{\alpha_a - \alpha_b}{\beta} \right) X'^{ab}_{ni} X'^{ba}_{-ni} + \cot\left(\frac{\alpha_a - \alpha_b}{2} \right) \right]. \tag{A.27}$$

The force due to the Fourier modes X'_{ni} is given by

$$\begin{aligned} -\left(\frac{\partial H}{\partial X'_{ni}} \right)_{ab} = & \sum_{a \neq b} \left[\sum_{n=-\Lambda}^{\Lambda} \frac{N}{\beta} \left(n\omega - \frac{\alpha_a - \alpha_b}{\beta} \right)^2 X'^{ba}_{-ni} \right] \\ & + \sum_{a \neq b} \left[\sum_{n=-\Lambda}^{\Lambda} \left([X_{mj}, [X_{oi}, X_{-kj}]]_{ba} + [X_{-kj}, [X_{oi}, X_{mj}]]_{ba} \right) \right]. \end{aligned} \tag{A.28}$$

A problem in implementing the non-lattice simulation is the fact that we have to keep track of the sum of the Fourier modes, which requires a computationally heavy algorithm. This is why it is advisable to use parallel programming in this case.

The Fourier modes of the full model are given as

$$X_i^{ab} = \sum_{n=-\Lambda}^{n=\Lambda} X'^{ab}_{in} e^{\imath n\omega t}. \tag{A.29}$$

$$\Psi_\alpha^{ab} = \sum_{n=-\lambda}^{n=\lambda} \Psi'^{ab}_{\alpha r} e^{\imath r\omega t}. \tag{A.30}$$

Since r takes half-integer values and due to the anti-periodic boundary conditions in case of fermions, we have

$$\lambda \equiv \Lambda - \frac{1}{2}, \tag{A.31}$$

where λ is momentum cutoff for the Fermionic Fourier modes.

So, the full action can be written as

$$\begin{aligned} S = & \frac{N}{2\lambda} \sum_{n=-\Lambda}^{\Lambda} \sum_{a,k=1}^{N} \left(n\omega - \frac{\alpha_a - \alpha_b}{\beta} \right)^2 \beta X'^{ab}_{ni} X'^{ba}_{-ni} - \frac{N\beta}{4\lambda} \mathrm{Tr}\left(\sum_{k=-\Lambda}^{\Lambda} [X'_{ni}, X'_{mj}][X'_{oi}, X'_{-ki}] \right) \\ & + \frac{N}{2\lambda} \sum_{r=-\lambda}^{\lambda} \sum_{a,k=1}^{N} \imath \left(r\omega - \frac{\alpha_a - \alpha_b}{\beta} \right) \beta \bar{\Psi}'^{ba}_{\alpha r} \Psi'^{ab}_{\alpha r} - \frac{N\beta}{2\lambda} (\gamma_i)_{\alpha\beta} \mathrm{Tr}\left(\sum_{k=-\lambda}^{\lambda} \bar{\Psi}'_{\alpha r} [X'_i, \Psi'_\beta] \right) \end{aligned} \tag{A.32}$$

with

$$k = n + m + o, p = -k. \tag{A.33}$$

Discretising the potential part of the fermionic action is straightforward. We have

$$S_f^{pot} = \sum_{a,b=0}^{N^2-1} \sum_{m,n=0}^{\lambda-1} \sum_{\alpha,\beta=1}^{16} \Psi_{m,a}^{\alpha} \mathscr{M}_{mn,\alpha,\beta,ab}^{pot} \Psi_{n,b}^{\beta}, \tag{A.34}$$

$$\mathscr{M}_{mn,\alpha,\beta,ab}^{pot} = \frac{N\beta}{2\lambda} f^{abc} (\gamma^i)_{\alpha\beta} X_n^{c,i} \delta_{n+m,0}, \tag{A.35}$$

$$\mathscr{M}_{mn,\alpha,\beta,ab} = \frac{N\beta}{2\lambda} \left[\imath \left(r\omega - \frac{\alpha_a - \alpha_b}{\beta} \right) \sum_{k,l=0}^{16} t_k^{ba} t_l^{ab} - f^{abc} (\gamma^i)_{\alpha\beta} X_n^{c,i} \delta_{n+m,0} \right]. \tag{A.36}$$

The fermionic action may be written in the form

$$S_f = \frac{1}{2} \mathscr{M}_{A\alpha r;B\delta s} \Psi'^{A}_{\alpha r} \Psi'^{B}_{\delta s}, \tag{A.37}$$

where we have expanded

$$\Psi'_{\alpha r} = \sum_{A=1}^{N^2} \Psi'^{A}_{\alpha r} t^A, \tag{A.38}$$

in terms of the $SU(N)$ generators t^A which are normalised as $\mathrm{Tr}(t^a t^b) = \delta^{ab}$. Here f^{abc} are the structure constants for the $SU(N)$ generators. Thus $\mathscr{M}_{a\alpha,b\beta}$ is a $16(N^2-1) \times 16(N^2-1)$ antisymmetric matrix.

Integration over the fermionic matrices results in a Pfaffian Pf$\mathscr{M}$, which is complex in general. For implementing the HMC in the case of fermions, we need to integrate out the fermions giving

$$S_{eff} = S_B - \log(\det \mathscr{M}), \tag{A.39}$$

$$S_F = \sum_{n,k} \bar{\Psi}'_n \mathscr{M}_{nk} \Psi'_k. \tag{A.40}$$

The phase of the Pfaffian will not pose any problems during our simulations as it is observed that it is real positive in the temperature regime studied.

Hence, we can replace

$$|\mathrm{Pf}\mathscr{M}| = \det(\mathscr{D}^{\frac{1}{4}}) \text{ where } \mathscr{D} = \mathscr{M}^{\dagger} \mathscr{M}. \tag{A.41}$$

In this formulation, the Wilson loop operator for the BFSS model is given by

$$W = \frac{1}{N} \mathrm{Tr}\ \exp\big(\imath \mathrm{diag}(\alpha_1, \ldots, \alpha_N) + \beta X'_{09}\big). \tag{A.42}$$

Here, β is the extent of space along the temporal direction or, in other words, the inverse temperature.

If one implements the concept of Fourier acceleration to reduce the auto-correlation time [Catterall 08], the Metropolis update is given as

$$\begin{aligned} X_k'(t+\delta t_k) &= X'k(t) + \delta t_k P_k'(t) + \frac{\delta t_k{}^2}{2} F_k'(t), \\ P_k'(t+\delta t_k) &= P_k'(t) + \frac{\delta t_k}{2}(F_k'(t) + F_T'k(t+\delta t_k). \end{aligned} \tag{A.43}$$

The accelerated time step δt_k is related to the original time step as $\delta t_k = a_k' \delta t$ where the Fourier acceleration is given by different terms for bosonic and fermionic steps

$$\begin{aligned} \delta t_{kB} &= \frac{\Delta t(m_{acc}+2)}{\sqrt{\sin^2(\frac{2\pi k}{L}) + (m_{acc} + 2\sin^2(\frac{\pi k}{L}))^2}}, \\ \delta t_{kF} &= \frac{\Delta t \sqrt{\sin^2(\frac{2\pi k}{L}) + (m_{acc} + 2\sin^2(\frac{\pi k}{L}))^2}}{(m_{acc}+2)}. \end{aligned} \tag{A.44}$$

For optimum values, the value of the parameter m_{acc} is close to the mass gap.

Bibliography

[Ambjørn 00] Jan Ambjørn, Jun Nishimura, Konstantinos N Anagnostopoulos, Wolfgang Bietenholz & Tomohiro Hotta. (2000) **Monte Carlo studies of the IIB matrix model at large N.** Journal of High Energy Physics, vol. 2000,no. 07, page 011011, Jul 2000.

[Anagnostopoulos 07] Konstantinos N Anagnostopoulos, Masanori Hanada, Jun Nishimura, and Shingo Takeuchi (2007) **Monte Carlo studies of supersymmetric matrix quantum mechanics with sixteen supercharges at finite temperature** [arxiv:0707.4454 [hep-th]]

[Banks 97] Tom Banks, Willy Fischler, Steven H Shenker & Leonard Susskind. (1997) **M theory as a matrix model: A conjecture.** Physical Review D, vol. 55, no. 8, page 5112, 1997.

[Catterall 08] Simon Catterall & Sergey Karamov (2001) **Testing a Fourier Accelerated Hybrid Monte Carlo algorithm**

[Catterall 08] Simon Catterall and Toby Wiseman (2008), **Black hole thermodynamics from simulations of lattice Yang-Mills theory** Physical Review D, vol. 78, no. 4, page 1502, 2008.

[Clarke 05] M A Clark & A D Kennedy.`https://github.com/mikeaclark/AlgRemez`, 2005.

[Drukker 99] Nadav Drukker, David J Gross & Hirosi Ooguri (1999)**Wilson Loops and Minimal Surfaces** Physical Review D, vol 60, 1999

[Filev 15] Veselin G Filev & Denjoe O'Connor (2015) **The BFSS Model on the Lattice** Journal of High Energy Physics, vol. 2016, no. 5, May 2016.

[Gattringer 07] Christof Gattringer & Christian B Lang. (2009) **Quantum Chromodynamics on the Lattice: An Introductory Presentation** (Lecture Notes in Physics Vol. 788), 2009.

[Hartle 03] James B. Hartle (2003) **Gravity - An Introduction to Einstein's General Relativity**. Addison-Wesley, 2003

[Hanada 07] Masanori Hanada, Jun Nishimura, & Shingo Takeuchi (2007) **Non-lattice simulation for supersymmetric gauge theories in one dimension**

[Itzhaki 00] Nissan Itzhaki, Juan M. Maldacena, Jacob Sonnenschein & Shimon Yankielowicz (2000)**Supergravity and The Large N Limit of Theories With Sixteen Supercharges**

[Jegerlehner 96] Beat Jegerlehner (1996) **Krylov space solvers for shifted linear systems** `http://arxiv.org/abs/hep-lat/9612014v1`

[Jevicki 99] Antal Jevicki, Yoichi Kazama and Tamiaki Yoneya (1999), **Generalized conformal symmetry in D-brane matrix models** Phys. Rev. D 59, 066001 1999

[Joseph 19] Anosh Joseph (2019) **Markov Chain Monte Carlo Methods: A Modern Primer** Springer `https://doi.org/10.1007/978-3-030-46044-0`

[Kabat 01a] Daniel Kabat, Gilad Lifschytz & David Lowe. (2001) **Black hole thermodynamics from calculations in strongly coupled gauge theory**. International Journal of Modern Physics A, vol. 16, no. 05, pages 856–865, 2001.

[Kabat 01b] Daniel Kabat, Gilad Lifschytz & David Lowe. (2001) **Black hole entropy from nonperturbative gauge theory**. Physical Review D, vol. 64, no. 12, page 124015, 2001.

[Kawahara 07] Naoyuki Kawahara, Jun Nishimura and Shingo Takeuchi, (2007) **Phase structure of matrix quantum mechanics at finite temperature** JHEP 0710, 097 (2007) [arXiv:0706.3517 [hep-th]].

[Klebanov 98] Igor R Klebanov & Leonard Susskind. (1998) **Schwarzschild black holes in various dimensions from matrix theory.** Physics Letters B, vol. 416, no. 1-2, pages 62-66, 1998.

[Maldacena 98] Juan Maldacena, (1998) **Wilson Loops in Large N Field Theories**, Phys. Rev.Lett. **80**, 4859 (1998);

[Metropolis 53] Nicholas Metropolis, Arianna W Rosenbluth, Marshall N Rosenbluth, Augusta H Teller & Edward Teller.(1953) **Equation of state calculations by fast computing machines.** The Journal of Chemical Physics, vol. 21, no. 6, pages 1087-1092, 1953.

[Nishimura 07] Jun Nishimura, Masanori Hanada, Shingo Takeuchi & Akitsugu Miwa (2007) **Schwarzschild radius from Monte Carlo calculation of the Wilson loop in supersymmetric matrix quantum mechanics** [arxiv:0811.2081 [hep-th]]

[Peskin 95] Michael E Peskin & Daniel V Schroeder (1995) **An Introduction to Quantum Field Theory** Boulder, CO

[Pestun 07] Vasily Pestun (2007), **Localization of Gauge Theory on a Four-Sphere and Supersymmetric Wilson Loops** ITEP-TH-41/07 PUTP-2248

[Polyakov 78] Alexander M Polyakov (1978) **Thermal properties of gauge fields and quark liberation**. Physics Letters B, vol. 72, no. 4, pages 477-480, 1978.

[Polchinski 98] Joseph Polchinski (1998) **String Theory (Vol 1) - An Introduction to Bosonic Strings**, Cambridge University Press

[Rey 98] Soo Jong Rey, Stefan Theisen & Jung-Tay Yee, (1998) **Wilson-Polyakov Loop at finite temperature in large-N gauge theory and anti-de Sitter supergravity** Nucl. Phys. B,527, 171, 1998

[Robert 04] Christian Robert & George Casella (2004). **Monte Carlo Statistical Methods** Springer-Verlag, New York.

[Schwarzschild 16] Karl Schwarzschild (1916), **Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin**, Phys-Math Klasse 1916, 189-196.

[Semenoff 04] Gordon W Semenoff (2004) **Matrix Model Thermodynamics** Quantum Theory and Symmetries

[Tanwar 20] Nikhil Tanwar (2020) **Monte Carlo Simulations of BFSS and IKKT Matrix Models**

[Ydri 17] Badis Ydri. (2017) **Computational Physics: An Introduction to Monte Carlo Simulations of Matrix Field Theory.** World Scientific, Singapore

[Zwiebach 04] Barton Zwiebach (2004) **A First Course in String Theory** Cambridge University Press